



780-968-8859
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OPERATIONS MANUAL
MODEL: ALC_B_V2_0

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Start, Stop, Reset

When Hand Off Auto Switch is selected in Hand the VFD will bypass the pump off controller, speed reference will be via the door mounted potentiometer, run command via Hand Off Auto Switch. All well shutdowns are active when in hand and in auto. When the Hand Off Auto Switch is selected in Auto HMI Main Screen will perform run commands and speed reference (see below).



Figure 1 ALC-B Interface

HMI Main Screen (Start/Stop/Reset)

The main screen of the POC's HMI features various buttons and indicators that offer limited feedback and control over the POC. A sample layout of the main screen is shown in the figure below. **Note: The SE200 will not start unless the HOA switch is set to the Auto position.**

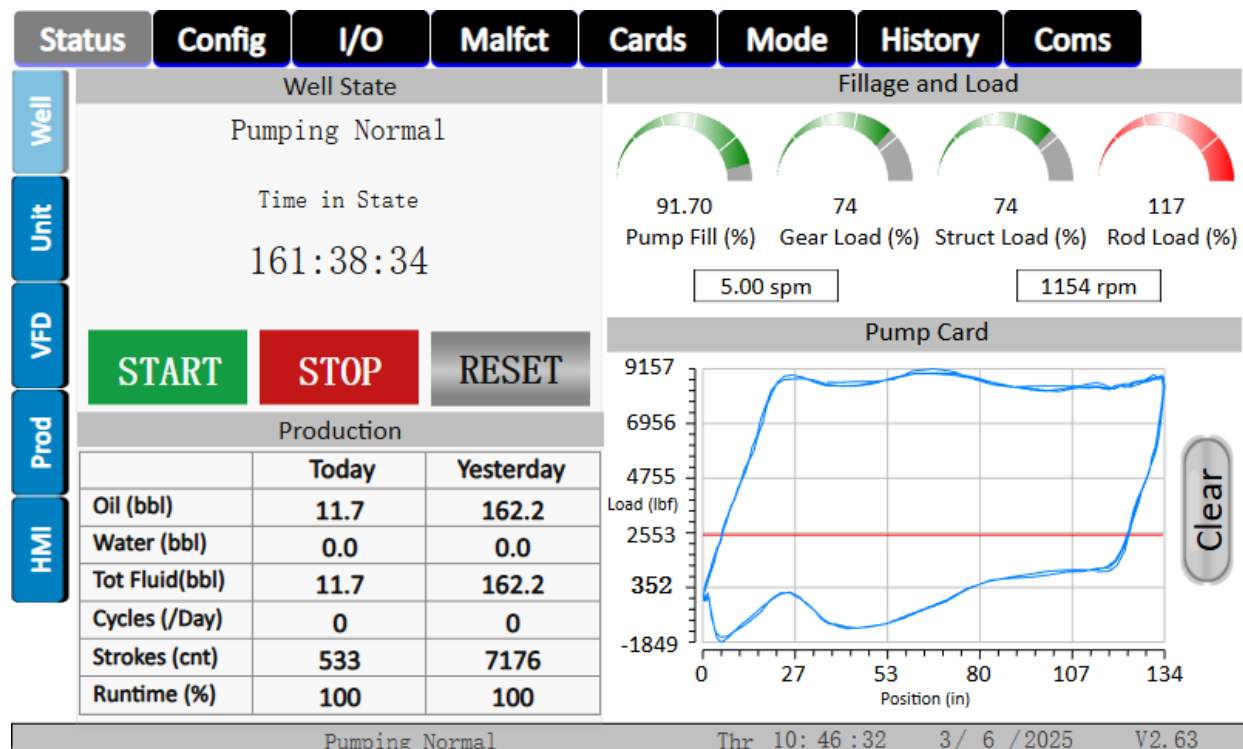


Figure 2 Main Screen

The following list describes the function of each button and indicator:

- **START** – The Start Pumping Unit button, when pressed, will signal the POC to start the pumping unit. If the unit is in a faulted state, the fault must be reset before the unit will be able to start.
- **STOP** – The Stop Pumping Unit button, when pressed, will signal the POC to stop the pumping unit.
- **RESET** – Pressing the Fault Reset button will clear any active fault on the POC.
- **Well State** – This indicator displays the current state of the POC.
- **Time in State** – This indicator shows the amount of time that the POC has been in the current state displayed in the Well State indicator. The format is HH:MM: SS, where H represents the digits for hours, M for minutes, and S for seconds.
- **Pump Fill (%)** – Displays the current pump fillage in %.
- **Gear Load (%)** – Displays the current gearbox load in % based off the type of pumping unit installed.
- **Struct Load (%)** – Displays the current structure load in % based off the max rated structure limit of the unit.
- **Rod Load (%)** - Displays the current rod load in % based off the max entered rod load limit.
- **Strokes per Minute** – Displays the calculated strokes per minute of the pumping unit.

- **RPM** – Displays the RPM that the unit is operating at.
- **Oil (bbl)** - Displays the total amount of oil produced in barrels. **(Note: these are estimated values and not exact numbers)**
- **Water (bbl)** – Displays the total amount of water produced in barrels. **(Note: these estimated values and not exact numbers.**
- **Total fluid (bbl)** – Displays the sum of the amount of fluid and oil in barrels. **(Note: these estimated values and not exact numbers.**
- **Cycles (/Day)** – Shows how many start/stop cycles were completed the previous day.
- **Runtime (%)** – Shows the percent runtime of the unit for the previous day.
- **Pump Fillage (%)** – Displays how full the pump is as a percentage of its total volume.
- **Pump Card** – Shows the current pump card being drawn, the clear button erases all drawn cards on chart.
- **Menu** – The menu navigation is divided into an X (horizontal) and Y (vertical) axis. Where every menu on the X axis has a different set of sub menus on its respective Y axis.

Pumping Unit Status

The Pumping Unit Status page provides several indicators for viewing the status of various portions of the pumping unit, listed below, and are shown in the following figure.

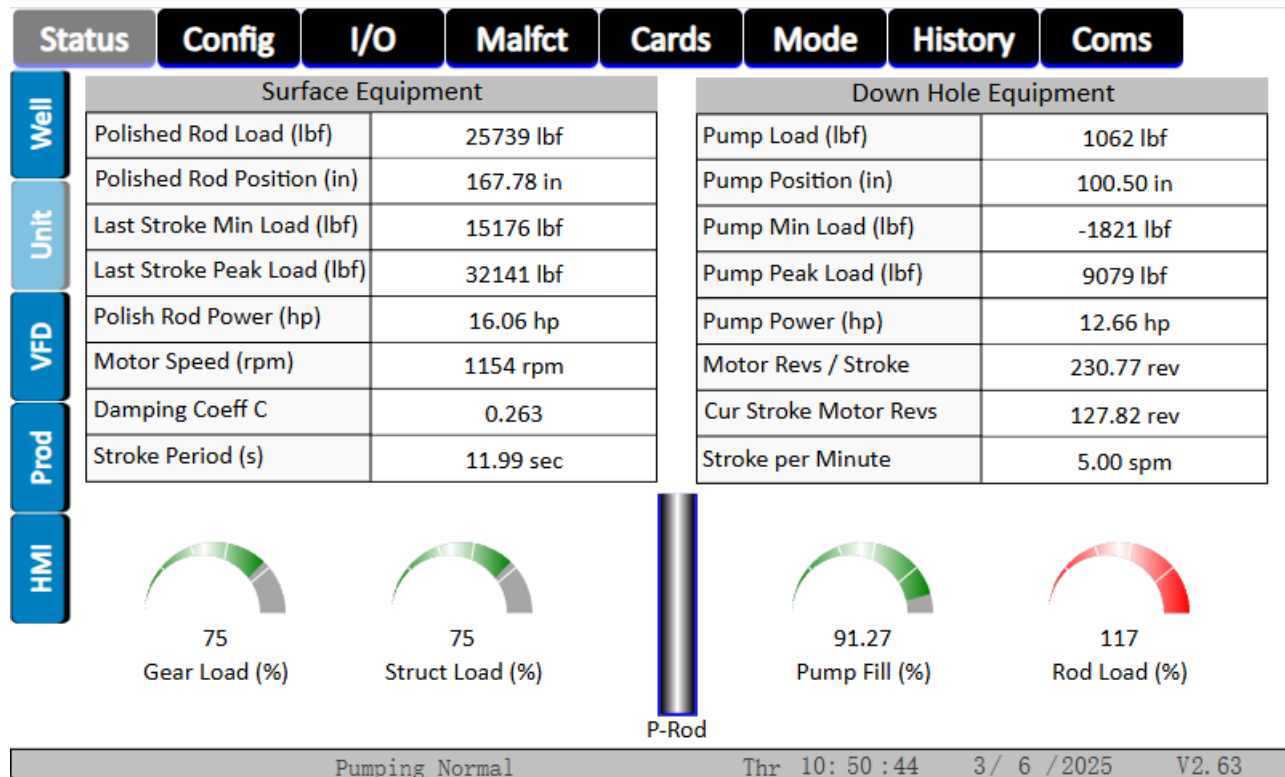


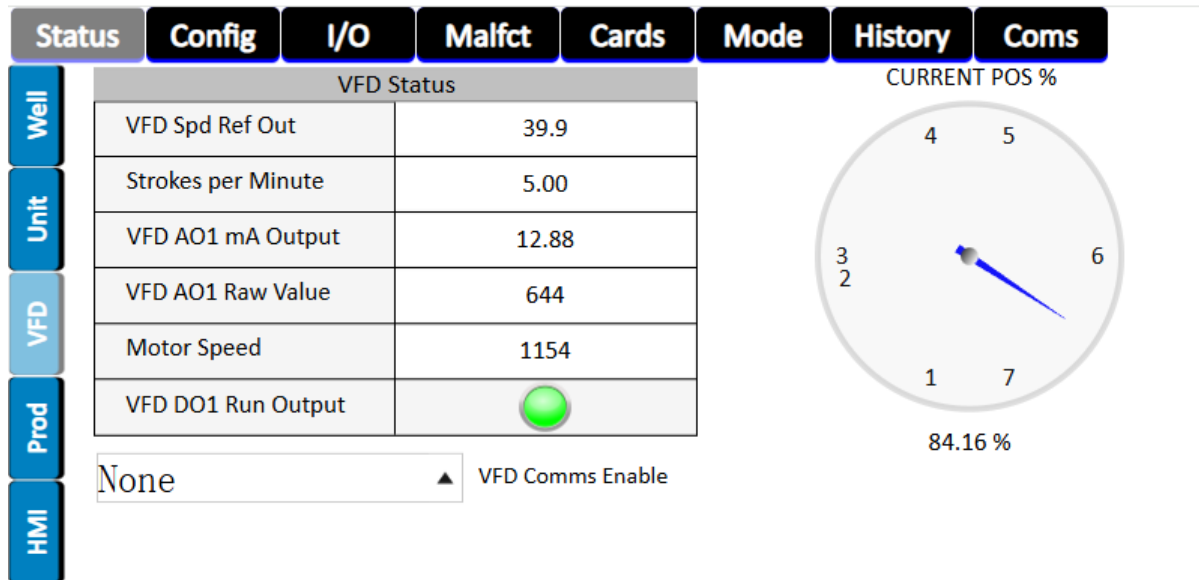
Figure 3 Pumping Unit Status

- **Polished Rod Load** – Displays the load, in pound force, on the polished rod.
- **Polished Rod Position** – Indicates the position, in inches, of the polished rod relative to the surface of the wellhead.
- **Last Stroke Min Load (lbf)** – Indicates the last stroke minimum rod load.
- **Last Stroke Peak Load (lbf)** – Indicates the last stroke peak rod load.
- **Polish Rod Power (hp)** – Indicates the calculated horsepower based off the surface card and is only updated once per stroke.
- **Motor Speed (rpm)** – Displays the current speed of the motor in RPM.
- **Damping Coeff (C)** – Indicates the calculated friction coefficient.
- **Stroke Period** – Displays how many seconds it takes the unit to complete one stroke.
- **Pump Load** – Shows the current load on the pump in pounds.
- **Pump Position** – Indicates the calculated position of the pump.
- **Pump Min Load (lbf)** – Indicates the last stroke minimum pump load.
- **Pump Peak Load (lbf)** – Indicates the last stroke pump peak load.
- **Pump Power** – Shows how much power, in horsepower, the rods are exhorting on the pump.

- **Motor Revs / Stroke** – Displays how many revolutions of the motor occur for each stroke of the unit.
- **Cur Stroke Motor Revs** – Displays how many revolutions of the motor that have occurred within the current stroke.
- **Stroke per Minute** – Displays how many strokes occur per minute.
- **Gear Load** – Calculated maximum load in % of reducer rating. Assuming perfectly balanced.
- **Structure Load** – Maximum load on polish rod as a % of the structure rating.
- **Pump Fill (%)** – Displays the current pump fillage in %.
- **Rod Load** – Calculated stress on the most loaded taper as a % of that tapers rating.

VFD Status

The VFD Status page provides several indicators for viewing the status of various portions of the BEPOC's control board with regards to its signals its sending to the VFD. See the image below of the page and its description listed below.



Pumping Normal Thr 10: 51 :35 3 / 6 /2025 V2.63

Figure 4 VFD Status

- **VFD Spd Ref Out (Hz)** – This displays the BEPOC's output speed reference to the VFD in Hz.
- **Strokes per Minute** – This displays the strokes per minute of the unit.
- **VFD AO1 mA Output (mA)** – This displays the mA output of the AO1 (VFD Speed Ref out).
- **VFD AO1 Raw Value** – This displays the raw value of the AO1 output.
- **Motor Speed (rpm)** – This displays the motor speed in rpm.
- **VFD DO1 Run Output** – This displays the output status of the BEPOC's VFD run output. When the green dot is bright green, it means the BEPOC is calling for the unit to run.
- **VFD Comms Enable** – This drop-down menu allows the user to enable comms via the selected port.
- **Current POS %** - Displays the current position and the corresponding percent speed based off the max output speed set by the user.

Production Status

The Inferred Production page is used for viewing the calculated production of POC for the current day and the previous day. An example of this page is shown in the figure below and the parameter descriptions are listed below it.

Status

Config

I/O

Malft

Cards

Mode

History

Coms

Well

Unit

VFD

Prod

HMI

Production			Pump Status	
	Today	Yesterday	Pmp Fillage(%)	91.88 %
Oil (bbl)	14.7 bbl	162.2 bbl	Net Stroke (in)	121 in
Water (bbl)	0.0 bbl	0.0 bbl	Gross Strk (in)	132 in
Tot Fluid(bbl)	14.7 bbl	162.2 bbl	Fluid Load (lbs)	9059 lbf
Cycles (/Day)	0 /day	0 /day	Pump Card Min Load (lbf)	-1745 lbf
Strokes (cnt)	667 cnt	7176 cnt	Pump Card Peak Load (lbf)	9003 lbf
Runtime (%)	100 %	100 %	Pump Intake Press (psi)	0.00 psi

Pumping Normal

Thr 11:13:16

3/ 6 /2025

V2.63

Figure 5 Inferred Production

- **Current Fluid** – Shows the amount of fluid pumped, in barrels, for the current day.
- **Current Oil** – Shows the amount of oil pumped, in barrels, for the current day.
- **Current Water** – Displays the amount of water pumped, in barrels, for the current day.
- **Current Strokes** – Displays the number of strokes that have occurred within the current day.
- **Current Cycles** – Indicates how many Start-Stop cycles the unit has gone through within the current day.
- **Current % Run** – Shows how long the pumping unit has been running, as a percentage, within the current day.
- **Yesterday Fluid** – Shows the amount of fluid pumped, in barrels, for the previous day.
- **Yesterday Oil** – Shows the amount of oil pumped, in barrels, for the previous day.
- **Yesterday Water** – Displays the amount of water pumped, in barrels, for the previous day.
- **Yesterday Strokes** – Displays the number of strokes that occurred the previous day
- **Yesterday Cycles** – Indicates how many Start-Stop cycles the unit went through the previous day.
- **Yesterday % Run** – Shows how long the pumping unit ran, as a percentage, during the previous day.
- **Pump Fillage (%)** – Displays how full the pump is with each stroke. It is shown as a percentage.
- **Net Stroke** – Displays the portion of the stroke that moved fluid, in inches.
- **Gross Stroke** – Shows the total displacement of the pump, in inches, for the current stroke.
- **Fluid Load** – Indicates the fluid load on the pump by first calculating the friction load then removing it from the pump card.
- **Pump Card Peak Load** – Displays the peak load, in pounds, that was on the pump during the last stroke.
- **Pump Card Min Load** – Shows the minimum load, in pounds, that was on the pump during the last stroke.
- **Pump Intake Press** – Displays the intake pressure, in psi, at the pump during the current stroke.

HMI Status

The HMI Status page provides the user with basic information about the HMI and POC’s software version and most recent reset time. There is also a QR code that when scanned provides the user with the Machine ID.

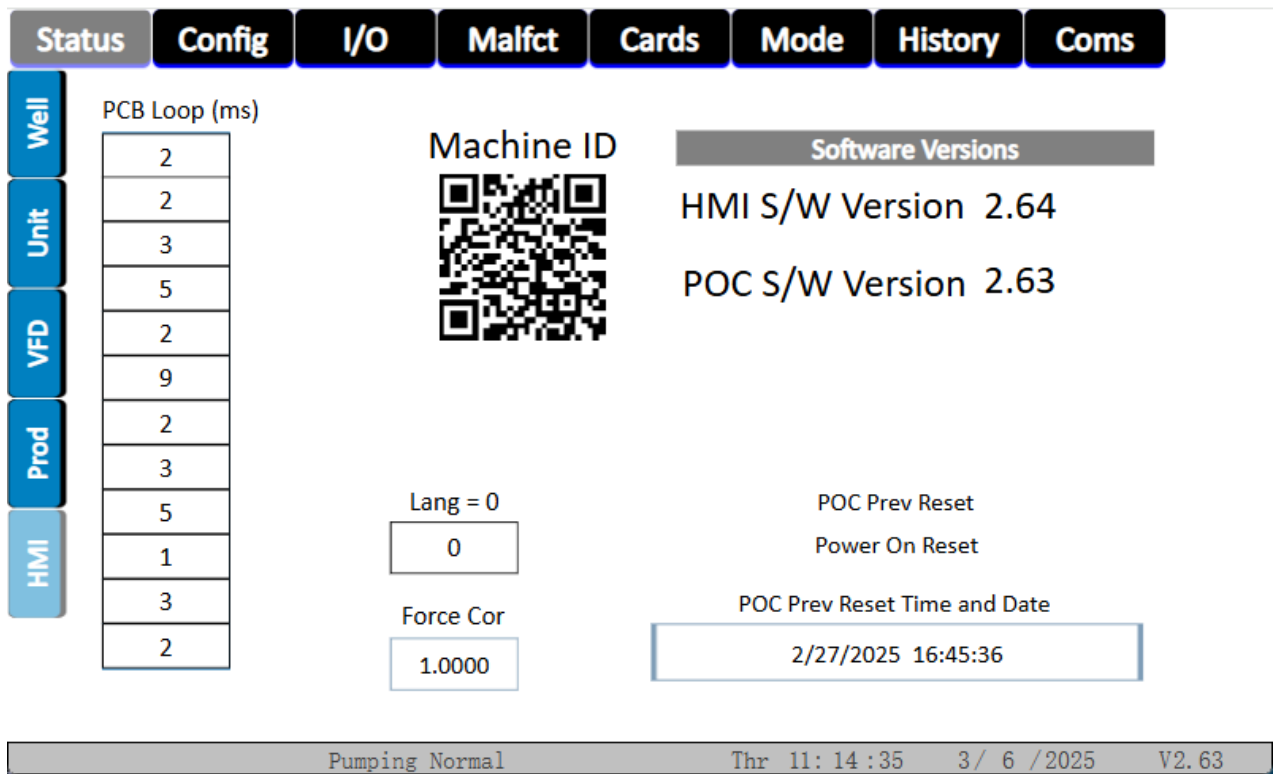


Figure 6 HMI Status Page

Well Config Page

The Rod Taper Set-up page provides the means to set up and modify up to six tapers for the rod string. As well as the pump information for the production status page. The page is split up into three columns; the first column is used to set the rod length for each taper in feet, the second column is used for setting the rod diameter for each taper in inches, and the third column is used to select whether a taper is made of steel or fiberglass. To edit a field, press it and enter or select a value. **NOTE:** Any unused tapers should have their rod length and rod diameter set to **zero**. The figure below provides an example taper setup. No 0-length taper is allowed between tapers. The **Buoyed Rod Weight** is the buoyant weight of the entire string in lbs. This weight should be within **500lbs** of your **Measured Rod Weight**.

Status		Config		I/O		Malft		Cards		Mode		History		Coms	
Well	Rod Tapers														
			Rod Length (ft)		Rod Dia. (in)		Rod Type		Production						
	Rod Taper 1		3000		1.00000		Steel		Pump Plunger Dia.	IPA K Factor					
	Rod Taper 2		3000		0.87500		Steel		1.50 in	1.00					
	Rod Taper 3		3000		0.75000		Steel		Gauge Off Hour (0-23)	Gauge Off Min (0-59)					
	Rod Taper 4		250		1.50000		Steel		9 hrs	0 min					
	Rod Taper 5		0		0.00000		Steel		Tubing Grad (psi/ft)	Tubing Head Press (psi)					
Rod Taper 6		0		0.00000		Steel		0.433 psi/ft	100 psi						
System					Buoyed Rod Weight (lb)		Measured Rod Weight (lb)		% Water Cut		Tubing Head Press Source				
	Restore Default Settings				19263		28096		0 %		Const.				
Adv.															
Pumping Normal Thr 11:15:5 3/6/2025 V2.63															

Figure 7 Rod Taper Set-up

When programming rod tapers as per rig report it's important that the technician note the following!

- **DO NOT RESTORE DEFAULTS UNLESS DIRECTED BY SEEL REP THIS WILL WIPE THE FACTORY SETTINGS!**
- Ensure proper measurement units are entered; the technician may need to convert metric to imperial based on units on rig report
- Do not count the polish rod or pony rod
- Each rod length is typically 25ft
- **DO NOT PUT MULTIPLE LINES OF THE SAME SIZE DIAMETER OF PIPE. ADD ALL LENGTHS TOGETHER AND PUT THEM IN THE SAME ROD TAPER LINE.**

The following parameters are optional and are used for the calculated production page.

- **Pump Plunger Diam** – Used for entering the pump plunger diameter in inches.
- **IPA K Factor** – used to adjust the calculations for pump plunger slippage along with any fluid volume shrinkage that occurs as gas separates from the solution in the production tank.
- **Gauge Off Hour** – This control is used to set the hour at which the values for the current day's production will be rolled over into Yesterday's production and the current day's production numbers will start over from zero. This number is based off a 24-hour clock, meaning that the only valid values for this control are between 0 and 23.
- **Gauge Off Min** – In addition to the Gauge Off Hour, this controls the minute at which the production roles over for the new day. Valid values for this control are between 0 and 59.
- **Tubing Grad** – This control is used to enter the force exerted by the weight of the fluid on the pump, in psi per foot. This number varies based on entrained gas and water content and is only used for calculating pump intake pressure.
- **Tubing Head Press** – This controls the constant value for the average tubing head pressure, in psi, and is only used for calculating the pump intake pressure.
- **Tubing Head Press Source** – Controls where the POC monitors the tubing head pressure. When Const. is selected, the POC uses the value in the Tubing Head Press control. When Analog is selected, the POC uses the scaled value from analog channel 1.
- **% Water Cut** – Controls the percentage of water in the fluid.

Below is an example of a well report in this example we will break down how to interpret the data and calculate it in a way that we are able to enter it into the POC. See the boxes highlighted below in red on where we will get this data.

For the **rod length**, we must calculate the total length of all **22.2mm rods**:

1. Multiply the total number of rods by their length in meters:
76.20 m + 586.74 m = 662.94 m
2. Convert meters to feet: **662.94 m x 3.28 = 2174.4'**
3. Now that the values are confirmed, enter **2175'** in the **Rod Taper #1** field under **rod length**
4. Next, we convert **22.2mm to inches**, which equals **0.875 inches**, and enter this value into the **pipe diameter** box for **Rod Taper #1**
5. Next will enter in the rod type **MMS stand for steel**

Since there is only **one remaining rod thickness**, we will enter the rest under **Rod Taper #2**.

1. Multiply the total number of rods by their length in meters:
160.02 m + 358.14 m + 381.00 m = 899.16 m
2. Convert meters to feet: **899.16 m x 3.28 = 2949'**
3. Now that the values are confirmed, enter **2949'** in the **Rod Taper #2** field under **rod length**
4. Next, we convert **19.1mm to inches**, which equals **0.750 inches**, and enter this value into the **pipe diameter** box for **Rod Taper #2**
5. Next will enter in the rod type **MMS stand for steel**

Final Step: Inputting the Pump Plunger Diameter

Based on the **model number description (25-200-RWBC-24-3-0-0HPT)**:

- 200 = **2"** thin-wall bottom hold-down
- 24 = **24'** barrel
- 3 = **3'** plunger
- HPT = **hollow pull tube**

From this, we enter **2 inches** into the **pump plunger diameter** field.

If the pump model number does not clearly indicate the pump size, consult the **service rig contact** for further verification.

See below for copy of rig report. See the boxes highlighted in **red** on where we get this data.

ALL mKB	WELL NAME:		H2S:		0.00%									
	PREPARED BY :		DATE :		LICENCE :		0							
	UWI:		Surface :		784.50		0							
	TD		KB Elev.		778.80		KB to TH							
	PBTD		Ground Elev.				KB to CF							
	CASING		SIZE (mm)		WEIGHT (kg/m)		GRADE		DEPTHS (mKB)					
	Surface Casing		219.10		35.72		J-55		610.00					
	Production Csg		139.7/114.3mm		34.23/22.47		L80/L80		1623.89/5137.8					
	Tubing		73.00		9.67		J55		1,618.26					
	Perf Date		Zone		Status		Gun Data/ Hole ID		Top		Bottom			
				Active		-		1,690.00		5,146.00				
SFC Csg 610.0	BOTTOM HOLE ASSEMBLY:													
	ITEM		DESCRIPTION						LENGTH (m)		TOP AT (m KB)			
			KBTHF						6		0.00			
			6,000daN Tension						0.32		6.00			
	1		101mm x 73.0mm tubing hanger						0.2		6.32			
	2		161 joints 73.0mm x 9.67kg/m J-55 Eue tubing						1560.87		6.52			
	3		139.7mm x 73.0mm Tryton "QTA" RHS/RHR tubing anchor w/22000 daN shea						1.4		1,567.39			
	4		2 joints 73.0mm x 9.67kg/m J-55 Eue tubing						19.88		1,568.79			
	5		73.0mm x 57.9mm friction seat "PSN"						0.33		1,588.67			
	6		1 - 73.0mm J-55 Eue tubing pup joint						1.22		1,589.00			
KOP 1,192.0	7		73.0mm x 60.3mm cross-over						0.24		1,590.22			
	8		88.9mm Lufkin Sprint Gas Separator						9.37		1,590.46			
	9		60.3mm x 88.9mm cross-over						0.1		1,599.83			
	10		2 - 88.9mm L-80 Eue 13.84kg/m tubing joints						19.31		1,599.93			
	11		88.9mm SHC/bull plug						0.24		1,619.24			
			Tubing bottom								1,619.48			
	WEIGHT OF TUBING STRING (daN)				TOTAL STRING LENGTH				1,613.48					
	TENSION/COMPRESSION (+/- daN)				STRETCH/SLACKOFF (+/- m)									
	LANDED STRING WEIGHT (daN)				TUBING BOTTOM (mKB)				1,619.48					
	Frac Interval: Top Perf		5114.6		4622.2		4129.3		3577.6		2943.3		2185.6	
			5079.9		4580.4		4073.6		3510.2		2876.8		2118.9	
			5037.8		4524.5		4018.2		3456.2		2809.2		2039.4	
			4983.4		4468.9		3975.9		3388.5		2729.2		1958.2	
			4941.3		4428.3		3921.2		3319.3		2648.8		1891.3	
			4885.3		4375.2		3866.7		3263.8		2581.7		1810.3	
			4830.9		4320.4		3824.8		3195.7		2502.2		1756.6	
4775.7			4279.5		3769.5		3129.8		2422.3		1690.0			
4733.2		4225		3700.1		3076.2		2344.2						
4677.8		4171.2		3644.6		3009.9		2263.6						
	DOWNHOLE DESCRIPTION FROM BOTTOM UP (Rod String)													
	AMOUNT		DESCRIPTION						LENGTH		TOTAL (m)			
	10		22.2 mm 6 per 2.5" LGXW scraper MMS sucker rods.						76.20		1580.40			
	50		19.1 mm 6 per 2.5" LGXW scraper MMS sucker rods.						381.00		1504.20			
	47		19.1 mm Plain MMS sucker rods.						358.14		1123.20			
	21		19.1 mm 6 per 2.5" STD scraper MMS sucker rods.						160.02		765.06			
	77		22.2 mm 6 per 2.5" STD scraper MMS sucker rods.						586.74		605.04			
	1		31.8 mm x 10.97m N-50 polish rod.						10.97		16.50			
	PUMP DESCRIPTION						PUMPJACK DESCRIPTION							
	BHP (LGP-9595), 25-200-RWBC-24-3-0-0HPT						Redhead 912-365-192 at 4SPM (Aug 2024 settings).							
	Max stroke - 235"													
	WELL HEAD DESCRIPTION													
	ITEM #		DESCRIPTION				SERIAL #		MWP		SIZE			
	1		NCE Casing Bowl				#CPI-E06954 (142319-45)		3k		9"			
	2		NCE Tubing Head				#T22-156-01		10k		5-1/8"			
	3		NCE Left Casing Valve				#210527047		10k		2-1/16"			
	4		NCE Right Casing Valve				#210527038		10k		2-1/16"			
	5		NCE Lower Master Valve				#2102261038		5k		2-9/16"			
6		NCE Upper Master Valve												
	REMARKS													

Figure 8 Example of well report

Pumping Unit Configuration

Figure 9 Pumping Unit Configuration

The Pumping Unit Selection page is used for setting up the POC to work with the correct pumping unit. This page includes five buttons and seven indicators which are listed below:

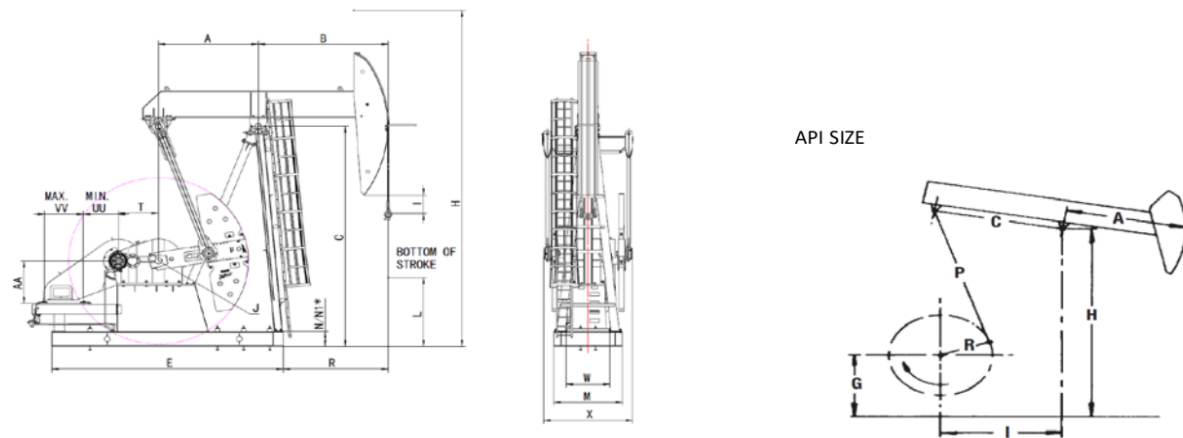
- **Unit API Size** – The use of this button is described under the subheading *Unit API Size* below.
- **Set Custom Unit** – Pressing this button will allow the user to set up a custom pumping unit, so long as the user has the **API Unit Dimensions** for the unit. To set a custom unit, press this button, the **Unit API Size** control will display the text “Custom Unit”, and enter the appropriate unit dimensions into the **API Unit Dimension** controls on the right on the screen. Please note that the values for the previous unit will remain in the dimension controls once this button is pressed.
- **Crank Hole #** – Use this button to indicate which of the crank arm hole that the pitman is connected to. Hole one will be the farthest from the gear box shaft, which would produce the longest stroke possible.
- **Pump Plunger Diam** – Used for entering the pump plunger diameter in inches.
- **Calculated Stroke** – This indicator displays the calculated stroke, in inches, of the pumping unit.
- **Unit Type** – This button is used for the selection of the unit type. Options will be auto selected based on the API size selected but can be changed. The options are:
 - Conventional
 - Mark II
 - Reverse Mark
 - Torque Master
 - Air Balance
 - Beam Balance

- **Unit Rotation** – Use this dropdown menu button to select the direction of rotation for the crank arm. The two options are Clockwise and Counterclockwise. The rotation is determined by observing the unit with the wellhead on your right and watching the rotation of the crank arm as the unit is operating.
- **Factory Defaults for all Parameters** – This button is used for restoring the POC to its factory default settings and can be used to reset the controller in cases where too many settings have been changed, and it would be easier to start over from factory defaults then it would be to trace all the incorrect settings. This is also useful on initial startup of the unit, to bring the settings to a known state.

The following can only be set in custom unit selection:

- **R Dimension** – The radius of crank arm in inches determined by crank hole number.
- **K Dimension** – The fixed distance between the crank arm center and the walking beam fixed pivot point.
- **C Dimension** – The distance between the walking beam fixed pivot point and the pitman arm connection to walking beam.
- **P Dimension** – The length of the pitman arm.
- **A Dimension** – Distance from front of horse's head to the walking beam fixed pivot point.
- **I Dimension** – Horizontal Distance from the crank arm center to the walking beam fixed pivot point.
- **Gear Reducer** – The API Reducer # or rating of reducer in (in-lbs)/1000 ex: 456,000 is a 456.
- **Structure Rating** – The API Structure # or rating of structure in (lbs/100) ex: 30,500 is a 305.
- **Structural Unbal** – The structural unbalance in lbs of the unit for gear reducer load calculations.
- **Phase Angle** – The phase angle of the unit in degrees for gear reducer load calculations.

DIMENSIONAL DATA



Dimensional Data (in.)

A	B	C	E	H	I	J	L	M	N	R	T	UU	VV	AA	X	W
137.80	285.43	410.74	281.73	669.38	19.10	125.20	56.20	88.58	35.43	209.83	66.54	36.93	45.13	61.02	133.45	56.77

API Geometry Dimensions (in.)

A	C	P	R	H	G	I
285.43	137.98	281.89	73.03	410.74	127.95	137.80

General Technical Data

Type			NC-2560-500-320		Unit
Max Load Rating			50,000		lbs.
Stroke Length			320 276 234 193		in.
Crankshaft Rated Torque			2,560,000		in.-lbs
Stroke Frequency			1 ~ 8		SPM.
Phase Angle			0		degrees
Gear Reducer	Gear Type		DOUBLE REDUCTION INVOLUTE GEAR REDUCER		/
	Gear Ratio		28.29		/
	Shaft Center Distance		66.54 [High speed 25.59 , Low Speed 40.94]		in.
	Max Speed		11		rpm
	Sheave	Option 1	68"-8-8V	☐	
		Option 2		☐	
		Option 3		☐	
		Option 4		☐	
		Option 5		☐	
	Crank shaft diameter		11.81		in.
	Gearbox reducer oil capacity		220		gallons
Oil in gear reducer		AGMA ISO VG 220 EP#5 for operation down to -18 °C AGMA ISO VG 150 EP#4 for operation bellow -18°C down to -30 °C		/	
Max Noise Level			< 92		Db.(A)
Max Vibration Velocity			< 0.25		in./s
Torque Factor	@ 90° / Max. Stroke		148.49		in.
	@ 270° / Max. Stroke		-154.72		in.
Structural Unbalance			-5130		lbs
ECB	Cranks Only / Max. Stroke		-880		lbs
Counterweights × Number			14500*4		lbs.
Max ECB			30856		lbs
Wireline Type & Hanger			Ø36 ×16" CENTERS		/
Bolts Grade			SAE J429 Grade 5		/
Working Ambient Temp.			-30~-120		F
Bearing Life	Center bearing	Cylindrical roller bearings(single row)	Bearing load ratio R1= 2.45, L10 life = 116322 @8SPM		hours
	Equalizer bearing	spherical roller bearing	Bearing load ratio R1= 3.14, L10 life = 232961 @8SPM		hours
	Crank pin bearing	Spherical roller bearing	Bearing load ratio R1= 3.10, L10 life = 90397 @8SPM		hours
Bearing grease specification			AGMA No.7 (414~506 cSt. at 40°C) for operation down to -18 °C AGMA No.5 (198~245 cSt. at 40°C) for operation bellow -18°C down to -30 °C		

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Figure 10 Example Pumping unit API dimension

Unit API Size

When the unit API size button is selected, a pop-up window appears on the display with a fifteen-unit segment of the complete list of pumping units stored on the POC, as shown in the figure below. To display a different set of units in the window, press either of the **Up** or **Down** buttons. To select a pumping unit, press on its unit size in the list, ensure that the correct section was made in the Unit API Size indicator, and press on the **X** in the upper righthand corner of the window or press **EXIT**. Pressing the **X** in the upper righthand corner or the **EXIT** button without making a unit selection will cancel the selection process and leave the previous unit API size.

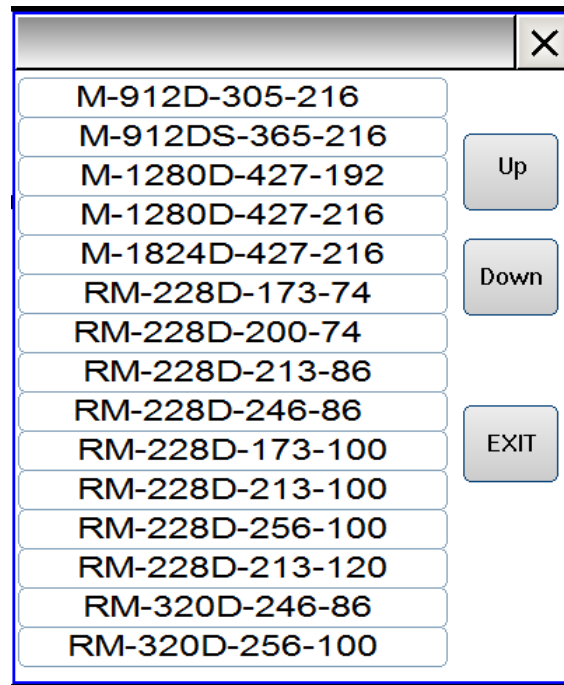


Figure 11 Pumping Unit Selection Page with Unit API Size window displayed

VFD Configuration

The VFD Configuration page is used for setting up the POC to speed up and slow down the drive when the Control Mode dropdown menu on the *Pump Off Selection and Set-Up* page is placed in either the VFD Surface or VFD Downhole modes. It is also used to set the VFD speed reference scaling for analog output 1 no matter what Control Mode is selected. The **VFD Min Spd Scale**, **VFD Max Spd Scale**, and **VFD Speed Ref Output Mode** are used to match the VFD analog input to the BEPOC analog output. The values in the BEPOC should match the values in the VFD. The **VFD Min Op Spd** and **VFD Max Op Spd** are used to set the operating speed range that the BEPOC is allowed to send to the VFD. The settings must meet the following criteria:

$$\text{VFD Min Spd Scale} \leq \text{VFD Min Op Spd} < \text{VFD Max Op Spd} \leq \text{VFD Max Spd Scale}.$$

Typical setup of the VFD speed scale

VFD Min Speed set to 0hz (this needs to match 0rpm that needs to be programmed into P0133 on the VFD)

VFD Max Speed set to 60hz (this needs to match 1200rpm that needs to be programmed into P0134 on the VFD)

VFD Output Speed Ref:

Set to 4-20mA (this needs to match the AI2 input programmed in the VFD, also need to ensure the dip switches on the VFD are set for 4-20mA for AI2, P0238)

If the values between the VFD HMI and the ALC-B HMI are not exact the gain can be adjusted in P0237

VFD Min Output Speed: as per desired

VFD Max Output Speed: as per desired

Pump Fillage Setpoint: as per desired

Deadband Pump Fillage: leave at default of 5% unless a different range is required

VFD Speed Increase%: leave at default of 5% unless a different range is required

VFD Speed Decrease%: leave at default of 1% unless a different range is required

Constant Speed: adjust if required via speed vs position window

The screenshot displays the VFD Configuration page with the following settings and controls:

- Tabs:** Status, Config (selected), I/O, Malfct, Cards, Mode, History, Coms.
- Left Sidebar:** Well, Unit, VFD, Position, System, Adv.
- Configuration Settings:**
 - Pump Fillage SP:** 90
 - Fillage Dead Band:** 5.00
 - VFD Speed Inc %:** 5
 - VFD Speed Dec %:** 1
 - VFD Speed Ref Output Mode:** 4-20mA
 - VFD Min Spd Scale:** 15.0
 - VFD Max Spd Scale:** 60.0
 - VFD Min Op Spd:** 20.0
 - VFD Max Op Spd:** 40.0
 - VFD Spd Ref:** 39.9
 - Spd Ref mA:** 12.88
 - SPM:** 5.00
 - VFD Speed Load Limit Set Point (lbs):** 60000
 - VFD Speed Load Limit Gain(%FS/lbs):** 0.00 (0 = Disabled)
- Control Mode:** Pump Card
- Const Speed %:** 100.0
- Speed Mode:** Const Spd
- Notes:**
 - <<< VFD Min Spd Scale and VFD Max Spd Scale should match the internal VFD Ref Scale Settings
 - If Control Mode is set to VFD Downhole: The Pump Fillage SP is used on the Downhole Card with the Dead Band in % fillage to increase or decrease VFD speed by the percentage of speed span set by VFD Speed Inc % and VFD Speed Dec %
 - See Speed vs Position Settings
- Status Bar:** Pumping Normal, Thr 11:16:10, 3/6/2025, V2.63

Figure 12 VFD Configuration

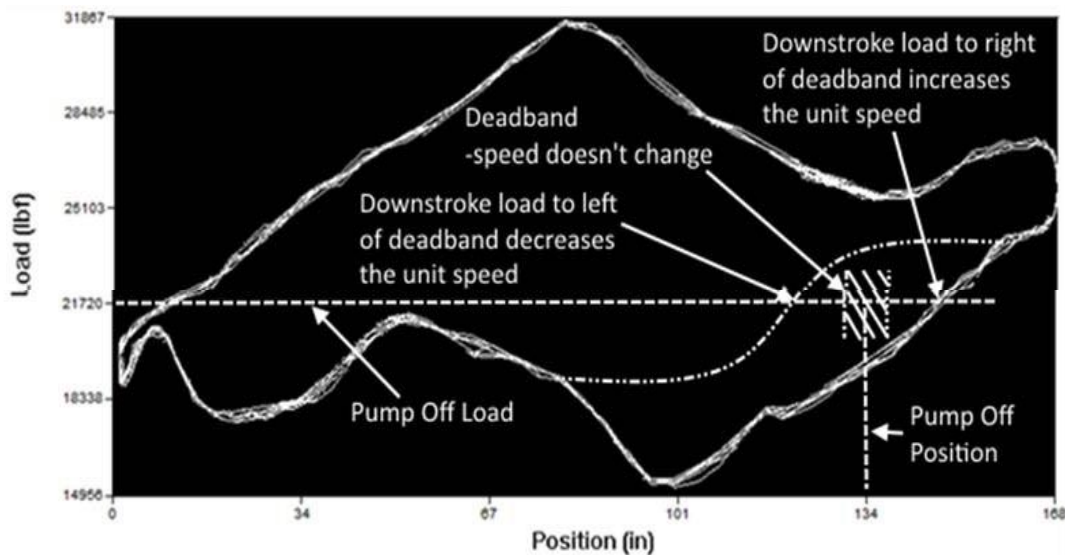


Figure 13 Deadband Description for VFD Surface mode

If the **Control Mode** is set to *VFD Surface*, a dead band will be calculated for the down stroke of the unit, based off the **Pump Off Position** and **Pump Off Load** and the value for the **Dead Band of Pump Fillage** control. The dead band is centered on the Pump Off Position along the Pump Off Load and extends to $\pm$ the Dead Band of Pump Fillage value. For example, using Figures 20 and 21, the Pump Off Load is 21720 lbs, the Pump Off Position is 134 inches, and the Dead Band of Pump Fillage is 5.00 inches, so the dead band starts at 129 inches and advances to 139 inches. When the down stroke of the pump card is outside the upper bound of the dead band (139 inches in the above example), the POC will begin stepping up the speed of the unit by the value in **VFD Speed Inc %** (5 % in this example, which corresponds to 5% of the Op Speed Span) to bring the load's position within the dead band. Once the down stroke load enters the dead band, the POC will stop increasing the unit's speed and instead hold it constant. When the load position extends below the bottom portion of the dead band (129 inches in this example), the POC will begin counting pump off strokes and slow down the unit in increments set by the **VFD Speed Dec %** (1 % in this example, which corresponds to 1% of the Op Speed Span). The speed will continue to decrease until the unit either reaches the maximum number of pump off strokes and shuts down, or the down stroke load position returns to the permissible range within the dead band, at which point the POC will keep the unit at a constant speed.

When the **Control Mode** is set to *VFD Downhole* with the **Shut Down Enables** set to *Pump Off Set Point*, the POC operates in the same manner as when Control Mode is set to VFD Surface, described in the previous paragraph, with the exception that it uses the downhole pump card instead of the surface card.

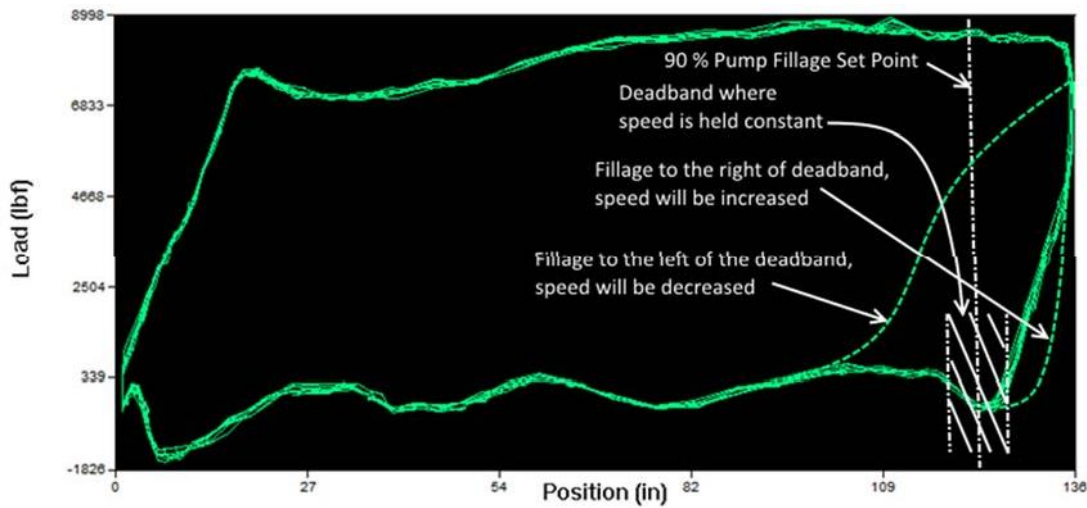


Figure 14 Deadband description for VFD Surface mode

If the **Control Mode** is set to *VFD Downhole* and the **Shut Down Enables** set to *Pump Fillage Set Point* (SP), a dead band will be calculated for the down stroke of the pump, based off the **Pump Fillage SP** and the value for the **Dead Band of Pump Fillage** control. The dead band is centered on the Pump Fillage SP and extends to $\pm$ the Dead Band of Pump Fillage value. For example, using Figures 20 and 22, the Pump Fillage SP is 90 with the Dead Band of Pump Fillage at 5.00, so the dead band starts at 85 % and advances to 95% of the pump fillage. When the down stroke of the pump card is outside the upper bound for the dead band (95 % in the above example), the POC will begin stepping up the speed of the unit by the value in VFD Speed Inc % (5 % in this example, which corresponds to 5% of the Op Speed Span) to bring the pump fillage within the dead band. Once the down stroke fillage enters the dead band, the POC will stop increasing the unit's speed and instead hold it constant. When the fillage extends below the bottom portion of the dead band (85 % in this example), the POC will begin counting pump off strokes and slow down the unit in increments set by the VFD Speed Dec % (1 % in this example, which corresponds to 1% of the Op Speed Span). The speed will continue to decrease until the unit either reaches the maximum number of pump off strokes and shuts down, or the down stroke fillage returns to the permissible range within the dead band, at which point the POC will keep the unit at a constant speed.

Speed vs. Position

The Speed vs. Position page is used in conjunction with a variable frequency drive to tune the speed of the pumping unit for various positions within a pumping stroke. Two modes are available for speed control, *Constant Speed* (Const Spd) and *Speed vs. Position* (Spd vs Pos), which are selected using the Speed Mode dropdown menu. When in constant speed mode, the drive will run the motor at the percentage of full speed set in the *Const Speed %* control. In speed vs. position mode, the speed of the pumping unit is allowed to vary based on the position of the unit within its stroke, with the Const Speed % used as an overall multiplier to achieve the desired strokes per minute (SPM). This page also includes an indicator for the percent displacement of the polished rod relative to the top of the stuffing box, an indicator for the current position of the pumping unit within its stroke (in percent), controls for the positions and speeds for the speed vs position mode, and indicators for the strokes per minute and the motor speed. An example screenshot of this page is shown in the figure below.

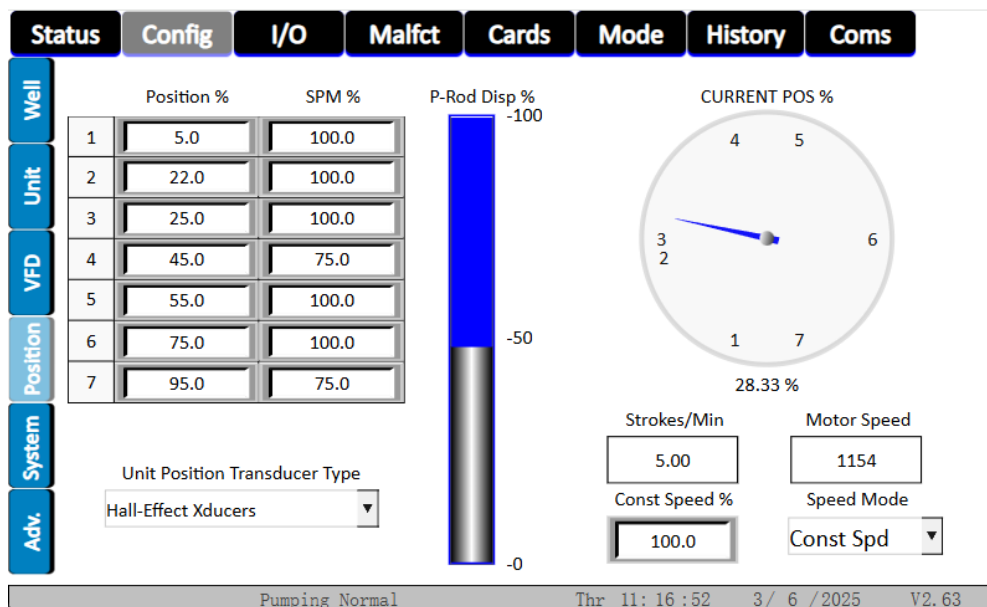


Figure 15 Speed Vs Position

For speed vs. position mode, each stroke is split up into seven different positions for the drive to speed up and slow down the pumping unit. The start of each position is placed in the Position % 1 through 7 controls. A visual representation for each of these positions can be seen on the pie chart overlapping the

CURRENT POS % clock-like indicator, with the colors for each slice displayed next to the corresponding Position % control. When entering a value for the start of a position, the value cannot be less than the value for the position ahead of it or greater than the value for the position that comes after it, otherwise the POC will autocorrect the value(s) to within the permissible range. Referring to the figure below as an example, if 4.0 is entered for Position % 2, the POC will move Position % 1 from 5.0 down to 3.9, but if

26.0 is entered for Position % 2, the POC will limit it to 24.9 to keep it from exceeding the value in Position % 3. The values in SPM % 1 through 7 control the percentage of full speed that the motor will run for the associated position within the stroke, with valid values being between 0 and 100%.

Time and Date

The Time and Date page is used for setting the 24-hour real time clock (RTC) on the POC and for synchronizing the HMI's RTC. To adjust the POC's month, day, year, etc., press on the desired field and enter in the desired value. To synchronize the HMI's RTC with the POC's RTC, press the **Sync HMI to POC RTC** button. **Note:** Pressing the *Sync HMI to POC RTC* button is only necessary on initial startup, as the HMI will synchronize its RTC with the POC's RTC every 83.3 minutes. The synchronization of the HMI to the POC is necessary to keep the timing of the alarms on the Alarms page in time with the alarm state on the POC.

The screenshot displays the 'Time and Date' configuration page. At the top, a navigation bar includes buttons for 'Status', 'Config', 'I/O', 'Malft', 'Cards', 'Mode', 'History', and 'Coms'. On the left, a vertical sidebar contains buttons for 'Well', 'Unit', 'VFD', 'Position', 'System', and 'Adv.'. The main content area is divided into two sections. The upper section, titled 'POC Real Time Clock Settings', is enclosed in a blue border and contains input fields for Month (3), Day (6), Year (2025), Hour (11), Minute (17), Second (25), and a Day of Week dropdown menu set to 'Thur'. Below these fields, it indicates '24 hour clock 0-23 for hour entry'. The lower section, titled 'HMI Clock is synchronized with POC Clock every 83.3 mins', is enclosed in a black border and contains input fields for Mon (3), Day (6), Year (2025), Hour (11), Min (17), Sec (25), and a W.Day dropdown menu set to 0. A 'Sync HMI to POC RTC' button is located to the left of the lower section. At the bottom, a status bar shows 'Pumping Normal', the current time 'Thr 11: 17 :25', the date '3 / 6 /2025', and the version 'V2.63'.

POC Real Time Clock Settings						
Month	Day	Year				
3	6	2025				
Hour	Minute	Second				
11	17	25				
24 hour clock 0-23 for hour entry		Day of Week				
		Thur				

HMI Clock is synchronized with POC Clock every 83.3 mins						
Mon	Day	Year				
3	6	2025				
Hour	Min	Sec				
11	17	25				
	W.Day					
	0					

Sync HMI to POC RTC

Pumping Normal Thr 11: 17 :25 3 / 6 /2025 V2.63

Figure 16 Time and Date

Advanced Settings

The Advanced Settings are used to rotate a pump card that is tilted up or down. First the **Pump Card Rotation Correction Enable** must be enabled. The amount of rotation required to level the card is the slope of the card either positive or negative. The slope can be calculated from the pump card page. The amount of load difference from the 0 position to the full stroke position divided by full stroke is the rotation (slope) correction required to give a level card. The formula would be (load difference) / (pump stroke length). The Pump Card offset can be used to push the pump card up or down by the load amount entered **Pump Card Rotation Correction Offset** in lbs. This can be a positive or negative number.

Status	Config	I/O	Malfct	Cards	Mode	History	Coms
Well	Pump Card Rotation Correction Enable ☐ OFF				Crank Arm Sensor Offset % 0.0		
Unit	Pump Card Rotation Correction lbf/inch 0						
VFD	Pump Card Rotation Correction Offset lbf 0				Unable to Stop Mode Disabled		
Position	Malfunction High SPM Enable ☐ OFF				Unable to Stop Time Secs 45		
System	Malfunction High SPM 25.00				Timed Mode Rod Load Limits Disabled		
Adv.	Analog Position Time Constant (ms) 0						

Pumping Normal Thr 11:17:51 3/6/2025 V2.63

Figure 17 Advanced Settings

- **Crank Arm Sensor Offset %** - is used to offset the crank arm hall effect sensor when it cannot be placed exactly at the bottom of pumping unit stroke. The user can set the offset from -50.0% to 50.0%. The offset is positive if the sensor is offset before reaching bottom of stroke and negative if the sensor is after reaching bottom of stroke. This is direction dependent. This should only be used when safety reasons don't allow placing the Hall Effect sensor exactly at bottom of stroke.

- **Unable to Stop Mode** – Selects the operation mode for the unable to stop function. When enabled the system will continue to monitor the well and draw all associated cards, the only way it would stop recording is based on receiving the pulse feedback on the following options:
 - **Disabled** – Disables the unable to stop function.
 - **Motor and Crank** – Enables the unable to stop function with the motor and crank arm sensors in conjunction to determine if the unit is stopped.
 - **Motor Only** – Enables the Unable to stop function with the motor sensor only to determine if the unit is stopped.
 - **Crank Only** – Enables the Unable to stop function with the crank arm sensor only to determine if the unit is stopped.
 - **Unable to Stop Time Secs** - Is used during the stopping state of the unit to give it time to come to a complete stop. The controller monitors the motor speed and/or crank arm DIs during the stopping state and if do DI is trigger for at least the last 10 secs of the “Unable to Stop Time” the controller will transition to the stopped state else it transitions to the Unable to Stop State. The minimum setting is 15 seconds, and the maximum setting is 300 seconds.
- **Malfunction High SPM** – When this function is enabled this will shut down the unit from exceeding a defined speed limit.
- **Analog Position Time Constant** – When using an inclinometer, this is letting you know how heavily filtered the analog signal is...the slower the unit is running the more filtering is required, it will overwrite automatically. (Ai7)
- **Time Mode Rod Load Limits** - This is to still have load protection and trending in Timed Mode Control, helps with diagnosis downhole while in such a simple control mode basis.

Analog I/O Status

The Analog I/O Status page contains two sets of indicators – one for the analog inputs and another for the analog outputs. Both sets contain two columns. For the analog inputs, the left-hand column displays the scaled input values, while the righthand column shows the raw input values for each analog channel. The scale factors for the analog inputs can be modified on the *Analog Input Config* page. For the analog outputs, the left-hand column displays the output value for each channel in milliamps, while the righthand column displays the raw analog value for each channel. The figure below provides an illustration of this page.

Status

Config

I/O

Malfct

Cards

Mode

History

Coms

AIO Stat

DIO Stat

AI Config

AI Fault

DI Fault

Internal

ANALOG INPUTS		
	Scaled Value	Raw Value
AI2	-24.95	5
AI3	-25.00	1
AI4	-25.00	0
AI5	-25.00	0
AI6	-25.00	0
AI7	3.23	2096

ANALOG OUTPUTS		
	mA Output	Raw Value
AO1	12.88	644
AO2	0.00	0
AO3	0.00	0
AO4	0.00	0
AO5	0.00	0
AO6	0.00	0

Pumping Normal

Thr 11:18:27

3 / 6 / 2025

V2.63

Figure 18 Analog I/O Status

Digital I/O Status

The Digital I/O Status page contains two sets of indicators – one for inputs and the other for outputs. Indicators that are blue show that their associated input or output is **OFF** while light green indicators show that their associated input or output is **ON**. As an example, in the figure below, Digital Input (DI) 1 is off, while DI9 is on.

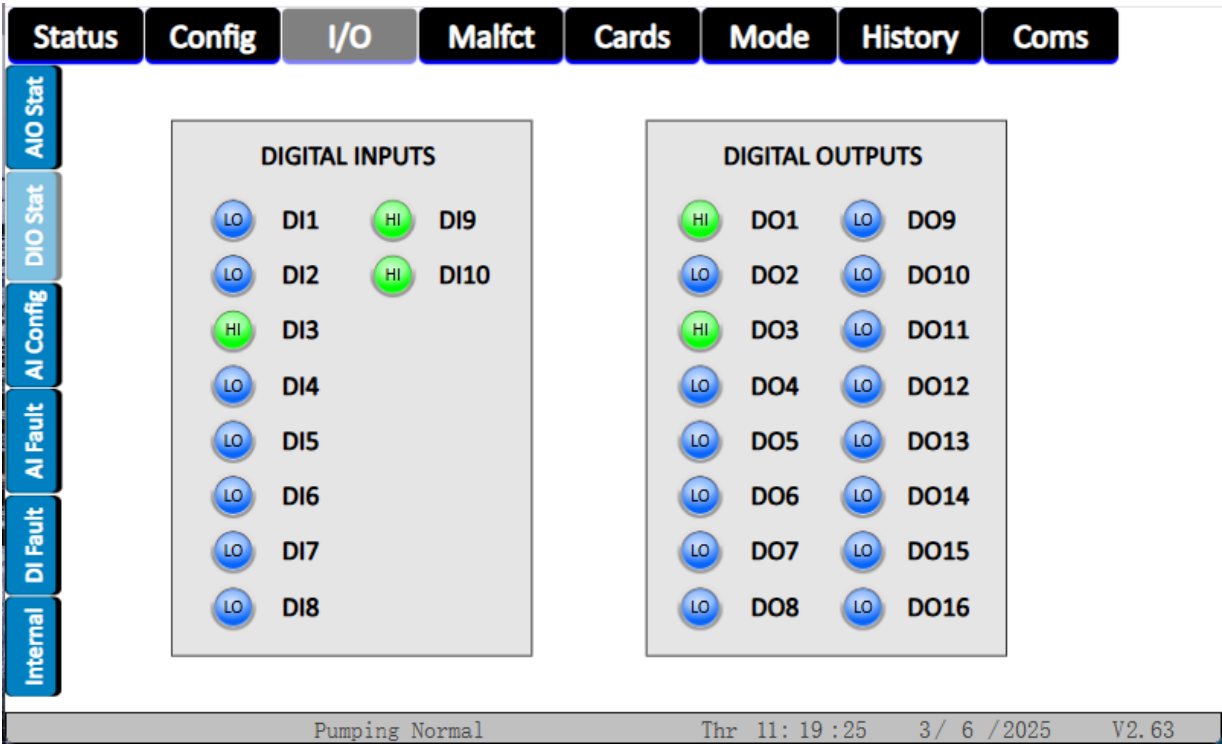


Figure 19 Digital I/O Status

Analog Input Configuration

The Analog Input Configuration page is used for setting the scaling of the analog values. The first column contains fields where the minimum scale value is entered; the second column contains fields for the maximum scale value contained; the third column is used for selecting the range of the input, via a dropdown menu; and the fourth column is meant for the selection of the units of measurement, also using a dropdown menu. Pressing the **Analog Fault Config** button will change the screen to the *Analog Input Fault Configuration* page. The **AI Filter Time Constant (ms)** is used to set the analog input lowpass filter time constant in milliseconds. The filter is applied to all 6 AI channels. If the filter time constant is set to 0ms or greater than 10000ms it is reset to the default 2000ms.

Status	Config	I/O	Malfct	Cards	Mode	History	Coms
AIO Stat DIO Stat AI Config AI Fault DI Fault Internal	Analog Input Scaling						
		Min Scale	Max Scale	Range	Units	Scaled Value	
	AI2	0.00	100.00	4 to 20 mA ▲	psi ▲	-24.96	
	AI3	0.00	100.00	4 to 20 mA ▲	psi ▲	-25.00	
	AI4	0.00	100.00	4 to 20 mA ▲	psi ▲	-25.00	
	AI5	0.00	100.00	4 to 20 mA ▲	psi ▲	-25.00	
	AI6	0.00	100.00	4 to 20 mA ▼	psi ▼	-25.00	
AI7	0.00	100.00	4 to 20 mA ▼	psi ▼	17.19		
Rod Load Cell Scaling							
	Max Weight	Gain	Offset Cnts	Polished Rod Load	AI Filter Time Constant (ms)		
AI8	50000 lbf	2	0	27398 lbf	2000		
Pumping Normal Thr 11:20:7 3/ 6 /2025 V2.63							

Figure 20 Analog Input Configuration

Rod Load Cell Scaling is also available from this page.

- **Rod Load Cell Max Weight** – Place the maximum rated weight for the load cell into this control. (Typically, 50000)
- **Rod Load Cell Gain** – Normally, this will be set to 1 but is used as a multiplication scale factor.
- **Rod Load Cell Offset Counts** – Normally, this will be set to 512 (2047 for ecowell units) but is used to correct an offset in the load cell.
- **Unit Position Transducer Type** – The default setting is “Hall-Effect Xducers” for the typical motor rpm and crank arm sensors to calculate unit position. Selecting “Analog Position Inclinator” allows for the use of an inclinometer on AI7 to calculate unit position. It is best if the inclinometer has a voltage output of at least 1 to 8 VDC or a max of 0 to 10VDC output.

Analog Input Fault Configuration

The Analog Input Fault Configuration page is used to set up the user-defined faults for each of the analog input channels AI2 – AI7 (one fault per channel). Using the AI Fault Configuration drop-down lists, the type of fault can be selected. The options for these controls are listed below:

Status	Config	I/O	Malfct	Cards	Mode	History	Coms
AIO Stat	Analog Input Fault Configuration						
	Scaled Value	Fault Level	Fault Reset Level	Fault Configuration			Status
DIO Stat	AI2 -24.96	0.00	0.00	Disabled ▲			✓
AI Config	AI3 -25.00	0.00	0.00	Disabled ▲			✓
AI Fault	AI4 -25.00	0.00	0.00	Disabled ▲			✓
DI Fault	AI5 -25.00	0.00	0.00	Disabled ▼			✓
Internal	AI6 -25.00	0.00	0.00	Disabled ▼			✓
	AI7 4.81	0.00	0.00	Disabled ▼			✓
Pumping Normal Thr 11:21:2 3/6/2025 V2.63							

Figure 21 Analog Input Fault Configuration page

- 1) **Disabled** – This disables the fault. When this is selected for a fault, the associated fault indicator at the bottom of the page will appear to be on, but the fault itself will not be active.
- 2) **Hi Level Fault** – Once the scaled value rises above the value specified in the Aix Fault Level control, the POC will shut the unit down and the fault will only be reset when the user presses the **Reset** button on either the *Start Stop Reset* page or the *Alarms* page.
- 3) **Lo Level Fault** – Once the scaled value drops below the value specified in the Aix Fault Level control, the POC will shut the unit down and the fault will only be reset when the user presses the **Reset** button on either the *Start Stop Reset* page or the *Alarms* page.
- 4) **Hi Level w/ Reset** – Once the scaled value rises above the value specified in the Aix Fault Level control, the POC will shut the unit down. The fault can be reset by pressing the **Reset** button on either the *Start Stop Reset* page or the *Alarms* page, or the fault will get reset automatically by the POC when the scaled value drops below the value set in the Aix Fault Reset Level control.
- 5) **Lo Level w/ Reset** – Once the scaled value drops below the value specified in the Aix Fault Level control, the POC will shut the unit down. The fault can be reset by pressing the **Reset** button on either the *Start Stop Reset* page or the *Alarms* page, or the fault will get reset automatically by the POC when the scaled value rises above the value set in the Aix Fault Reset Level control.

Fault indicators for each of the faults are given at the bottom of the page. A fault is active when the indicator has turned red and is inactive when the indicator is green.

Digital Fault Configuration

The Digital Fault Configuration page is used to set up the built-in faults (DI1-DI3) and user-defined faults (DI4-DI10) for each of the digital input channels DI4 – DI10 (one fault per channel). Using the DI Fault Configuration drop-down lists, the type of fault can be selected. The options for these controls are listed below:

Set DI1, DI2 & DI3 to Hi Level Fault to have them disabled (once closed the input will alarm, non-fail safe)

Set DI4, DI5, DI6 to Lo Level Fault (once opened the input will alarm, fail safe)

If auto restart is required set to Lo Level Fault with Reset

- If auto reset is required ensure that the auto restart parameters in the VFD are also programmed (refer to VFD start up guide P1029, P1030, P1031)

Digital Input Fault Configuration			
	State	Fault Configuration	Status
DI4	LO	Disabled	✓
DI5	LO	Disabled	✓
DI6	LO	Disabled	✓
DI7	LO	Disabled	✓
DI8	LO	Disabled	✓
DI9	HI	Disabled	✓
DI10	HI	Disabled	✓

Fixed Digital Input Fault Configuration			
	State	Fault Configuration	Description
DI1	LO	Hi Level Fault	Hi Pressure
DI2	LO	Hi Level Fault	Hi Vibration
DI3	HI	Lo Level Fault	Brake Switch

DOx-POC Faulted Output Select
Not Used

Mode: N.C.
DOx-Unable to Stop Fault Out Sel
Not Used

Pumping Normal Thr 11:22:5 3 / 6 / 2025 V2.63

Figure 22 Digital Input Fault Configuration

- 1) **Disabled** – This disables the fault. When this is selected for a fault, the associated fault indicator at the bottom of the page will appear to be on, but the fault itself will not be active.
- 2) **Hi Level Fault** – This forces the fault to be a Normally Open type of fault. Once the input goes high, the POC will shut the unit down and the fault will only be reset when the user presses the **Reset** button on either the *Start Stop Reset* page or the *Alarms* page.
- 3) **Lo Level Fault** – This forces the fault to be a Normally Closed type of fault. Once the input drops out, the POC will shut the unit down and the fault will only be reset when the user presses the **Reset** button on either the *Start Stop Reset* page or the *Alarms* page.
- 4) **Hi Level w/ Reset** – This forces the fault to be a Normally Open type of fault. Once the input goes high, the POC will shut the unit down. The fault will be reset automatically by the POC when input is no longer sensed on the DI.
- 5) **Lo Level w/ Reset** – This forces the fault to be a Normally Closed type of fault. Once the input drops out, the POC will shut the unit down. The fault will be reset automatically by the POC when input is once again sensed on the DI.

Status indicators for each DI are shown on the left of each *Dlx Fault Configuration* control, while Fault indicators for each of the faults is given at the bottom of the page. A DI is active when its status indicator has turned to bright green and is inactive when the indicator is a dark green. A fault is active when its fault indicator has turned red and is inactive when the indicator is green.

DOx-POC Faulted Output Select is used to turn on a digital output when the POC is down other than when in Downtime Pumpoff, Downtime Host, or Down Time Timed Mode. This allows the user to select DO5 – DO16.

DOx-Unable to Stop Fault Output Select is used to turn on (or off depending on the “Unable to Stop Mode Select”) a digital output when the POC is faulted on the “Unable to Stop Fault”. This allows the user to select DO5 – DO16.

Unable to Stop Fault Mode Select is used to make the associated Digital Output either NO-Normally Open or NC- Normally closed.

Internal I/O Status

The **Internal I/O Status** page is a valuable troubleshooting tool that provides real-time insights into the BEPOC control board's operation. This page displays various key parameters, including:

- **Analog Inputs:** AI1, AI7, AI8, AI9, AI10, and AI11
- **Raw Load Cell Value:** Essential for verifying accurate weight measurements
- **Voltage Outputs:**
 - **5VDC (AI9)** for the load cell output
 - **3.3VDC (AI10)** for the load cell signal return at full scale (50000lbs).
 - **3.6VDC** for the Memory Backup battery

These readings enable technicians to quickly diagnose issues, verify sensor functionality, and confirm power outputs. Additionally, the **Bridge A/D Offset** and **Gain Compensation** can be fine-tuned directly from this page using a dropdown menu, allowing for precise calibration of the system.

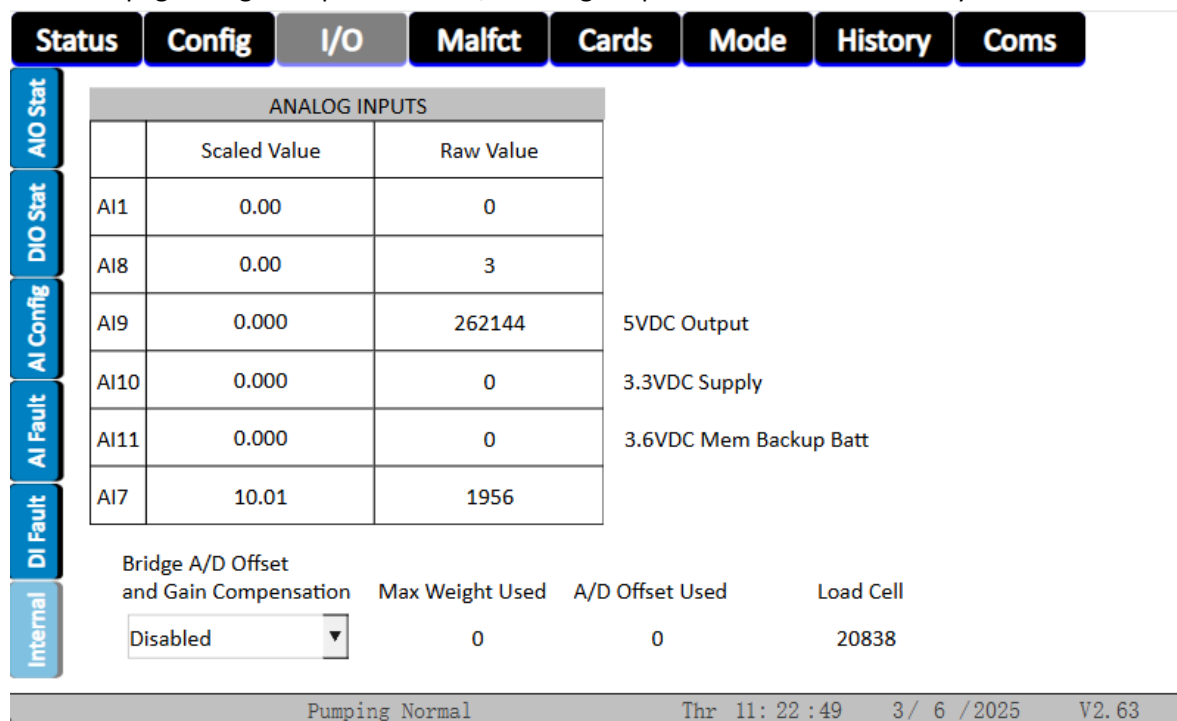


Figure 23 Internal I/O Status

Malfunction Min / Max Rod Load Settings

The Min/Max Rod Load Settings page, as seen in the figure below, is used to set the parameters necessary to shut down the pumping unit when it is carrying too light of loads or too great of loads. The list below describes the function for each of the buttons and indicators:

Ensure limits are within the rod limit and structure limit of the pumpjack.

Determine the running average rod load in ft/lbs once the unit has been running and an average load is established.

Status	Config	I/O	Malfct	Cards	Mode	History	Coms
Rod Load Last Stroke Min Load (lbs) 15571 Last Stroke Peak Load (lbs) 32337 Consec Strk Malf SP (cnt) 0 VFD Speed Load Limit Set Point (lbs) 60000 Min Load Limit (lbs) 2000 Peak Load Limit (lbs) 50000 Load Malf SP (lbs) 0 VFD Speed Load Limit Gain (%F.S./lbs) 0.00 Consecutive Min Load Allowed 2ms/Count 50 Consecutive Peak Load Allowed 2ms/Count 50 Consec Malf SP Strokes Allowed 0 0 = Disabled 0 = Disabled							
Pumping Normal Thr 11: 25 :52 3 / 6 /2025 V2.63							

Figure 24 Min / Max Rod Load Settings

- **Last Stroke Min Load** – This indicator displays the minimum load, in pounds, seen by the pumping unit on the last stroke.
- **Min Load Limit** – This control sets the minimum limit, in pounds, for loads being lifted by the pumping unit. Typical setpoint is 5000lbs less than the average down to 2500lbs.
- **Consecutive Min Load Allowed** – This control sets the maximum number of allowed minimum loads that the POC will tolerate before shutting down the pumping unit. The load is checked every two milliseconds by the POC, therefore the time to trip will be the count multiplied by two milliseconds. The typical setpoint is 100.
- **Last Stroke Peak Load** – This indicator displays the maximum load, in pounds, seen by the pumping unit during the last stroke.
- **Peak Load Limit** – This control sets the maximum limit, in pounds, for loads being lifted by the pumping unit. Typical setpoint max load set at 5000lbs greater than the average up to the structure limit of the pumpjack.
- **Consecutive Peak Load Allowed** – This control sets the maximum allowed peak loads that the POC will tolerate before shutting down the pumping unit. The load is checked every two milliseconds by the POC, therefore the time to trip will be the count multiplied by two milliseconds.

- **VFD Speed Load Limit Set Point** – Default is 60,000 lbs (disabled). Set this below the “**Peak Load Limit**” to have the VFD slow down if the present load is greater than the “**VFD Speed Load Limit Set Point**”. Typically set to 2000lbs less than the maximum load.
- **VFD Speed Load Limit Gain** – Default is 0 %/FS-lbs (disabled). This how aggressive the speed reduction is in % speed reduced per pound over the “**VFD Speed Load Limit Set Point**”. The typical setpoint is from 5-10%.
- **Consec Malf SP Strokes Allowed (See figure 24-A)** – When this parameter is set to any value greater than **0**, the unit will immediately switch to this mode. In this mode, the unit allows a specific number of consecutive strokes to exceed the **peak load limit** or fall below the **minimum load limit**, based on the **Load Malf SP (lbs)** parameter displayed to the left. Additionally, while operating in this mode, the **2ms/counts** setting no longer influences the shutdown process.
- **Load Malf SP (lbs) (See figure 24-A)** – The amount in pounds in which the unit isn’t allowed to exceed past its peak or minimum load limits. This number is the same for both.
- **Consec Malf Strokes (See figure 24-A)** – The amount of consecutive malfunction strokes in a row. After a stroke has fallen within the safe parameter limits the counter will reset to 0.

Consec Strk Malf SP (cnt) 0	Load Malf SP (lbs) 0	Consec Malf SP Strokes Allowed 0 0 = Disabled
---	------------------------------------	--

Figure 24-A Min / Max Rod Load Settings

Pump Card

The **Pump Card** page displays a graph that plots **Pump Cards** as they are captured by the POC. The graph dynamically adjusts its axes based on the maximum and minimum values of the plotted data.

Key Features:

- A **Clear** button resets the graph for new data.
- The **Well State** indicator at the bottom of the page displays the current unit status.
- Similarly, the pumping statistics page provides additional helpful information to the user.
- **Fill Base Settings** Fill base settings come into play usually when the jack is put to a shorter stroke. but the actual pump barrel is not changed, . This setting also helps with the fillage calculation, when the unit thinks the pump is fuller than it is due to DH drag. So essentially when the rods/tubing are lowered then this setting will be lowered slightly to adjust for actual fluid load vs DH friction, helping the controller pick between these.

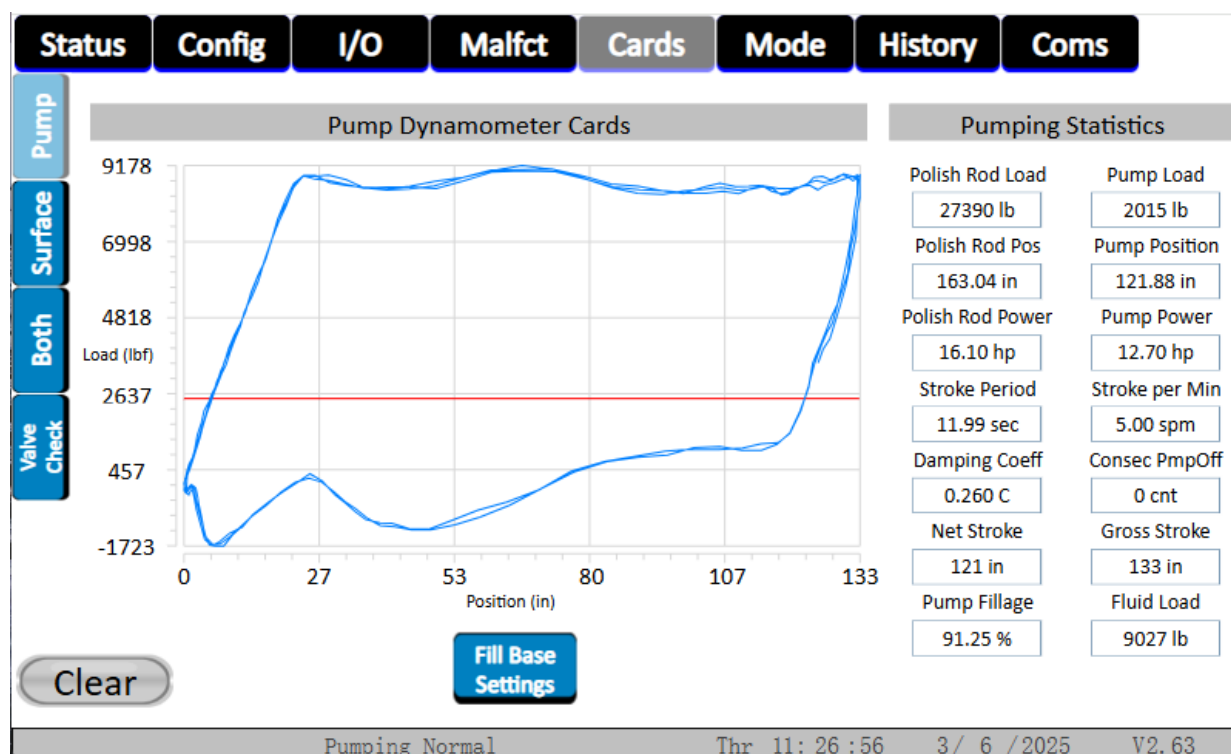


Figure 25 Pump Card

Surface Card

The **Surface Card** page displays a graph plotting **Surface Cards** as they are captured by the POC. The graph dynamically adjusts its axes based on the maximum and minimum values of the plotted data.

Key Features:

- A **Clear** button resets the graph, allowing new data to be displayed.
- The **Well State** indicator at the bottom of the page shows the status of the unit.
- Ability to adjust the **Pump Off Load** and **Pump Off Position** set points when running in **Pump Off Control Mode**.

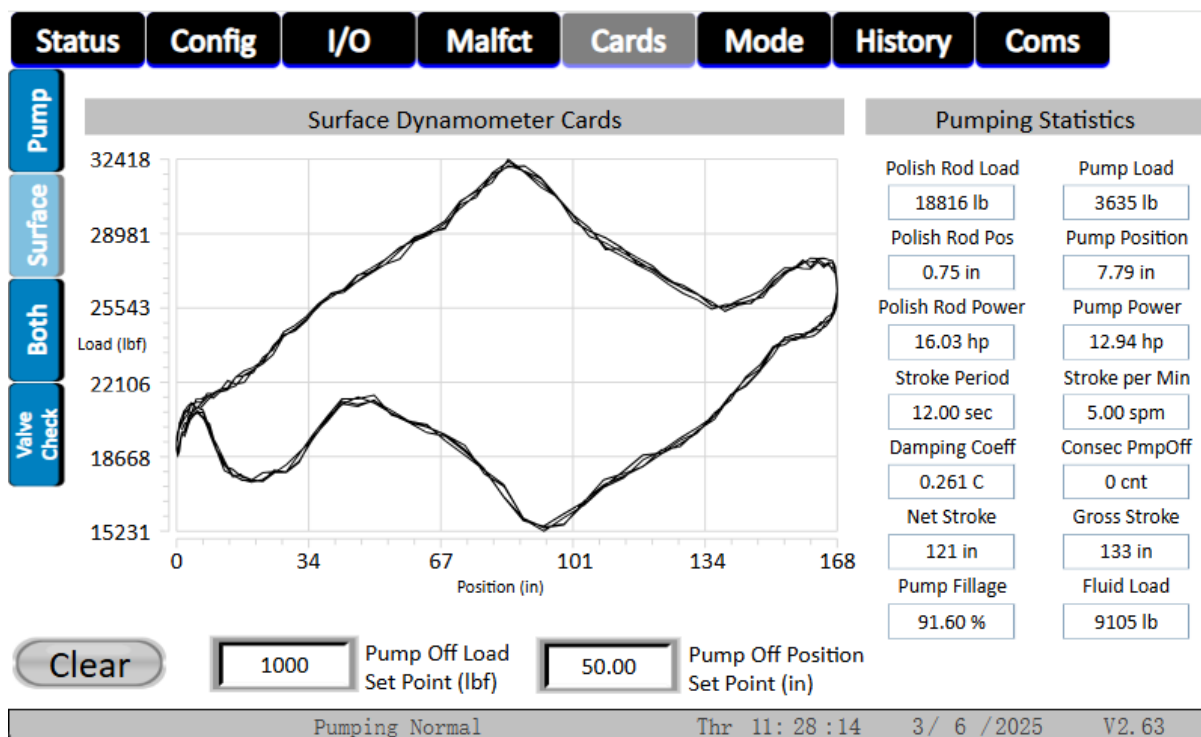


Figure 26 Surface Card

Surface and Pump Card

The **Surface and Pump Card** page displays a graph that plots **Surface Cards** and **Pump Cards** as they are captured by the POC.

Key Features:

- **Surface Card (Black Plot)** – Measured directly using unit instrumentation.
- **Pump Card (Blue Plot)** – Calculated based on the Surface Card and rod taper.

Additionally, two **red horizontal lines** represent operational load limits:

- **Peak Load Limit (Top Line)**
- **Min Load Limit (Bottom Line)**

If the Surface Card extends beyond either limit, the POC will **shut down** the pumping unit. These limits can be adjusted on the **Min/Max Rod Load Settings** page, which are accessible from this menu below the graph.

Additional Features:

- The **graph scales dynamically**, expanding or contracting based on the plotted data.
- A **Clear** button resets the graph for new data.
- The **Well State** indicator at the bottom of the page shows the current unit status.
- Depending on the **Control Mode** selected in the **Min/Max Rod Load Settings**, a **black or blue crosshair** will appear where the **Pump Off Load Set Point** and **Pump Off Position Set Point** intersect.
- The **Pumping Statistics** side menu displays additional useful information for the user.

For more details on setting up these points, refer to the **Pump Off Selection and Setup** page in this document.

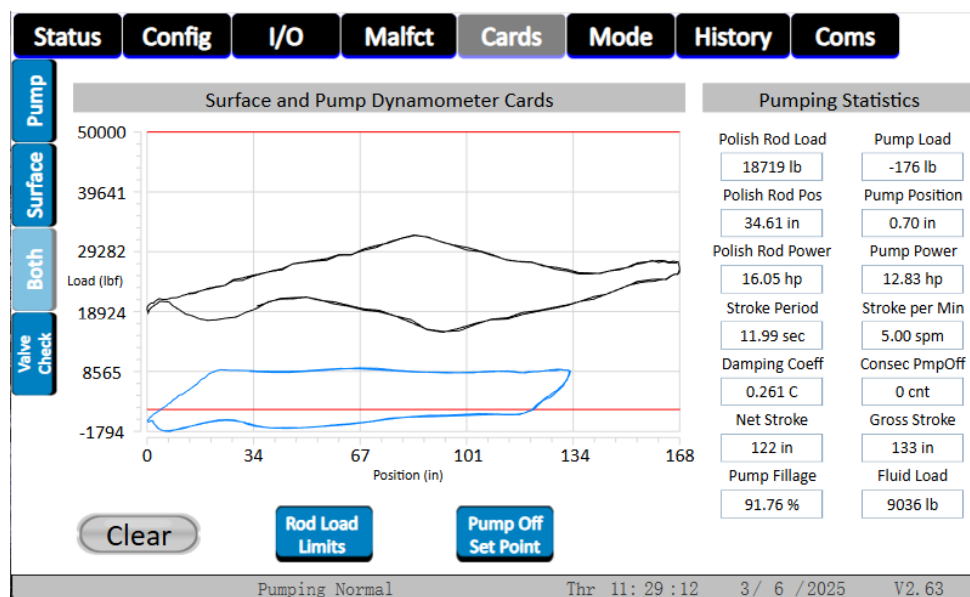


Figure 27 Surface and Pump card

Valve Check - Polish Rod Load vs Time

The Polish Rod Load vs Time page is used to aid the user in testing their traveling valve and standing valve, though the procedures for these tests are outside the scope of this manual. This page is only meant to give the user a feel for what is going on downhole and *does not* provide any calculated values that would be useful in the previously mentioned tests.

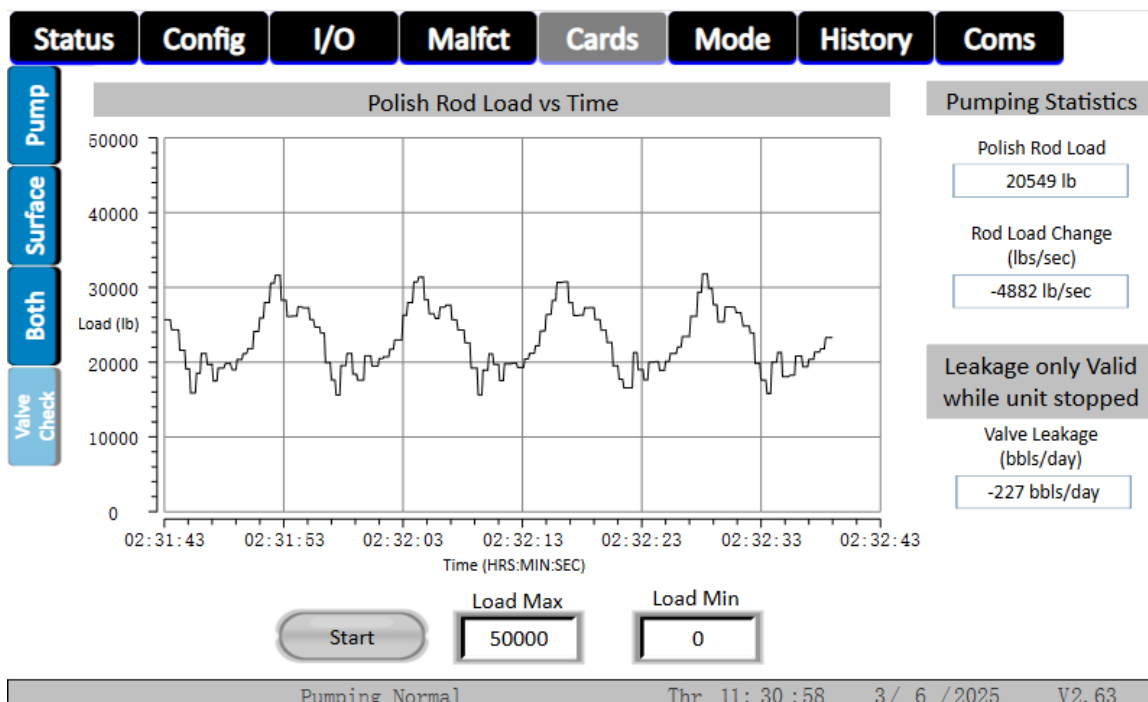


Figure 28 Polish Rod Load vs Time

To make the graph start plotting the polish rod load, press the Start button. This will cause the **Start** button to turn into the **Stop** button and pressing it again will stop the data collection. Pressing the **Clear** button will clear the graph, while setting the **Y Max** value will set the maximum polish rod load that will be displayed, and setting the **Y Min** value will set the minimum polish rod load to be displayed. The graph shows approximately one minute's worth of data.

When testing the traveling valve, stop the unit with the pumpjack at the top of the stroke (weights at the bottom).

- Test for 60 seconds
- If there is a decrease in the load this indicates that the traveling valve is not operating properly.

When testing the standing valve, stop the unit at the bottom of the stroke (weights at the top).

- Test for 60 seconds
- If there is an increase in load this indicates that the standing valve is not operating properly.

NOTE: In either scenario, further investigation is required.

Pump Off Selection and Set-up

The Pump Off Selection and Set-up page is used for setting the startup, pump off, and operation parameters on the POC. An example of this screen is shown in the figure below.

TYPICAL ALC-B SETUP

Shutdown Enables: Set unit for pump fillage setpoint Control Mode: VFD Downhole

Operation Mode: Normal Mode

Consecutive Pump off strokes allowed: As per desired Pump fillage setpoint: As per desired

FOR SEEL TESTING USE ONLY: Set motor rpm (1180 rpm) in oriented startup on VFD (re-confirm P 0133: 0rpm, P0134 - 1200rpm)

Status	Config	I/O	Malfct	Cards	Mode	History	Coms
PumpOff Linear Energy	Power On Delay (Seconds)		Start Alert (Seconds)		Minimum Pump Strokes		
	10		5		15		
	Control Mode		Operation Mode				
	Pump Card		Normal Mode				
	Pump Off Load Set Point (lbf)		Pump Off Position Set Point (in)		Consecutive Pump Off Strokes Allowed		
	1000		50.00		3		
	Pump Off Down Time (hrs)		Pump Off Down Time (mins)		Pump Fillage Set Point %		
	0		30		90.0000		
	Timed Mode On (hours)		Timed Mode On (mins)		Shut Down Enables		
	0		30		☐ OFF Pump Off Set Point ☒ ON Pump Fillage Set Point		
Timed Mode Off (hours)		Timed Mode Off (mins)					
0		30					

Pumping Normal Thr 11:31:41 3 / 6 / 2025 V2.63

Figure 29 Pump Off Selection and Set-up

The list below describes the controls and indicators in this page:

- **The Power On Delay** is used for delaying startup of the unit after a power outage. This can be used to stage the startup of multiple units.
- **The Start Alert** (If equipped) is how long the POC will sound the horn every time the unit starts.
- **The Minimum Pump Strokes** is used to set the minimum number of strokes that must occur before the pump/surface cards can be used for determining shutdown.
- **Control Mode** – The following options are provided for this control:
- **Surface Card** – When this option is selected, the POC examines the Pump Off Set Point on the unit's down stroke to see if it is below the surface card. If the set point is below the down stroke of the surface card, then the POC counts that stroke as a consecutive pump off stroke, incrementing the Consecutive Pump Off Stroke counter. Once this counter reaches the limit set in the Consecutive Pump Off Strokes Allowed control, the POC will shut the unit down. When the set point is in the area above the down stroke for the surface card, the consecutive pump off stroke counter is reset and the unit is allowed to continue operation. NOTE: When this mode is selected, the POC will not allow the Pump Fillage Set Point to be selected. Additionally, Pump Off Set Point must be selected for pump off

to occur.

- **Pump Card** When the Pump Card control mode and Pump Fillage Set Point are selected, the pump fillage will need to be above the Pump Fillage Set Point % or the POC will begin counting Consecutive Pump Off Strokes and shut the unit down once it has reached the consecutive number of pump off strokes allowed.

NOTE: When Pump Card is selected, the Shut Down Enables will automatically switch to Pump Fillage Set Point but can be changed to Pump Off Set Point. To do this, deselect Pump Fillage Set Point by pressing it, then select the Pump Off Set Point.

- **Not Used** – If this option is selected, then the POC will not be in a control mode for pump off.
- **VFD Surface** – please refer to the VFD Configuration page below.
- **VFD Downhole** – please refer to the VFD Configuration page below.
- **Operation Mode** – The options for this control are listed below:
- **Normal Mode** – This mode allows the unit to be run based on the mode selected in the Control Mode controller.
- **Timed Mode** – When selected, the POC will start and stop the unit based on the timers set up in the Timed Mode section of this page.
- **Host Mode** – When this option is selected, the POC will start and stop the unit based solely on the commands received from SCADA.
- **Pump Off Load Set Point** – This control sets the maximum load, in pound force, that the POC will allow before pumping off, in turn controlling the vertical movement of the pump off set point on the Surface and Pump Card page.
- **Pump Off Position Set Point** – Use this control to set the pump/surface card position, in inches, during the down stroke that the Pump Off Load Set Point will be used in determining pump off conditions. Modifying the value in this control allows for the horizontal movement of the Pump Off Set Point on the Surface and Pump Card page.
- **Consecutive Pump Off Strokes Allowed** – This controls the number of consecutive pump off strokes that will be allowed by the POC before pump off.
- **Pump Off Down Time (hrs)** – Controls the number of hours that the POC will remain in the off state after a pump off shutdown.
- **Pump Off Down Time (mins)** – Controls the number of minutes that the POC will remain in the off state after a pump off shutdown. The total down time will be the sum of both the Pump Off Down Time (hrs) and Pump Off Down Time (mins).
- **Pump Fillage Set Point %** - This controls the minimum percent fillage of the pump that the POC will allow before it starts counting consecutive pump off strokes in Pump Card and VFD Downhole modes while Pump Fillage Set Point is selected.
- **Timed Mode On (hrs)** – Controls the amount of time, in hours, that the POC will run the unit when in the Timed Mode of operation.
- **Timed Mode On (mins)** – Controls the amount of time, in minutes, that the POC will run the unit when in the Timed Mode of operation.
- **Timed Mode Off (hrs)** – Controls the amount of time, in hours, that the POC will keep the unit stopped after the Timed Mode On timer has expired in the Timed Mode of operation.
- **Timed Mode Off (mins)** – Controls the amount of time, in minutes, that the POC will keep the unit

stopped after the Timed Mode On timer has expired in the Timed Mode of operation.

Linear Pumping Unit Configuration

The Linear Pumping Unit Configuration starts with setting the **Unit API Size** to “Custom Unit” by pressing the **Set Custom Unit** button on the **Pumping Unit Selection** page then select “Linear Pumping Unit” from the **Unit Type** drop down on either the **Pumping Unit Selection** page or the **Linear Pumping Unit Configuration** page. Next set the **Distance Between Sprocket Centers** in inches. Then set the **Sprocket Radius** in inches. The system will calculate the stroke length from the two entered measurements.

If direction control (up stroke direction and down stroke direction) is required, the **Direction Control** set the **Sprocket Radius** to 0, set the **Distance Between Sprockets** to the pumping unit stroke length in inches. The POC will calculate position from a top of stroke proximity sensor, bottom of stroke proximity sensor, and time. An example would be a hydraulic pumping unit.

Direction Control is intended to use a position sensor for the top and bottom of stroke indication. This is useful for hydraulic or rack and pinion style pumping units. The direction control utilizes 2 digital outputs, one for upstroke and one for down stroke control. The **Jog Up** and **Jog Down** buttons are used to move the unit up or down when **Direction Control** is enabled, and the unit is in the **Stopped State**.

LINEAR PUMPING UNIT CONFIGURATION

Calculated Stroke (in)
167.99

Set Custom Unit

Distance Between Sprocket Centers (in)
0

Sprocket Radius (in)
0

Unit Type
Mark II

Const Speed %
100.0

Speed Mode
Const Spd

See Speed vs Postion Settings

Direction Control (Up and Down Stroke)
OFF

Jog Up

Jog Down

Pumping Normal Thr 11: 32 :18 3/ 6 /2025 V2.63

Figure 30 Linear Pumping Unit Configuration

Energy Management

The Energy Management page is used for setting two separate time periods for the controller to either shut down pumping operations or slow to VFD minimum speed settings. This feature also allows the operator to choose which days the feature will be active. The figure below provides an illustration of this page.

Figure 31 Energy Management

- EM Mode Select** – Selects the operating modes of the Energy Management function. The options are as follows:
 - Disabled - no action on any of the time periods.
 - Stop – Stops the pumping unit during the active time periods.
 - VFD Min Speed – Runs the pumping unit VFD at its “VFD Min Op Spd” setting found on the “VFD Configuration” page during the active time periods. This mode overrides all other VFD speed control functions.
- Monday – Sunday Enabled/Disabled buttons** – Selects which days the Energy Management function is active if the “EM Mode Select” is in “Stop” or “VFD Min Speed” mode.
- Energy Management Period 1** – Suspend Hour and Minute sets the desired time to begin either Stop or VFD minimum speed operation. Resume Hour and Minute sets the desired time to resume normal pumping operations. **Note:** this is a 24-clock setting i.e., 0-23 on the hour and 0-59 on the minute setting.
- Energy Management Period 2** – Suspend Hour and Minute sets the desired time to begin either Stop or VFD minimum speed operation. Resume Hour and Minute sets the desired time to resume normal pumping operations. **Note:** this is a 24-clock setting i.e., 0-23 on the hour and 0-59 on the minute setting.

Note: Either Time Period can be disabled by setting the Suspend Hour and Minute equal to the Resume Hour and Minute.

History Cards

The History Cards page provides access to several various cards stored on the POC. There are seven options to choose from – Stored cards, Shutdown 1 Cards, Shutdown 2 Cards, Standard Card, Start-up Card, Pump-Up Card, and Last Stroke. To select an option, press the **blue** arrows on either side of the card dropdown menu located beside the **Update** button, or select the pump cards directly from the card dropdown menu by pressing on it then pressing on the desired item in the list. Once the selection has been made, press the **Update** button and the HMI will produce the desired cards on the graph along with the timestamp for the cards above the **Update** button. The options for each card are described below:

- **Stored cards** – The last five surface and pump cards.
- **Shutdown 1 Cards** – The shutdown card plus the previous four cards.
- **Shutdown 2 Cards** – This is the previous shutdown card plus its four prior cards.
- **Start-up Card** – The card for the transition from starting state to the minimum strokes state.
- **Pump-Up Card** – The card for the transition from the minimum strokes state to the pumping normal state
- **Last Card** – This is the card for the previous stroke.

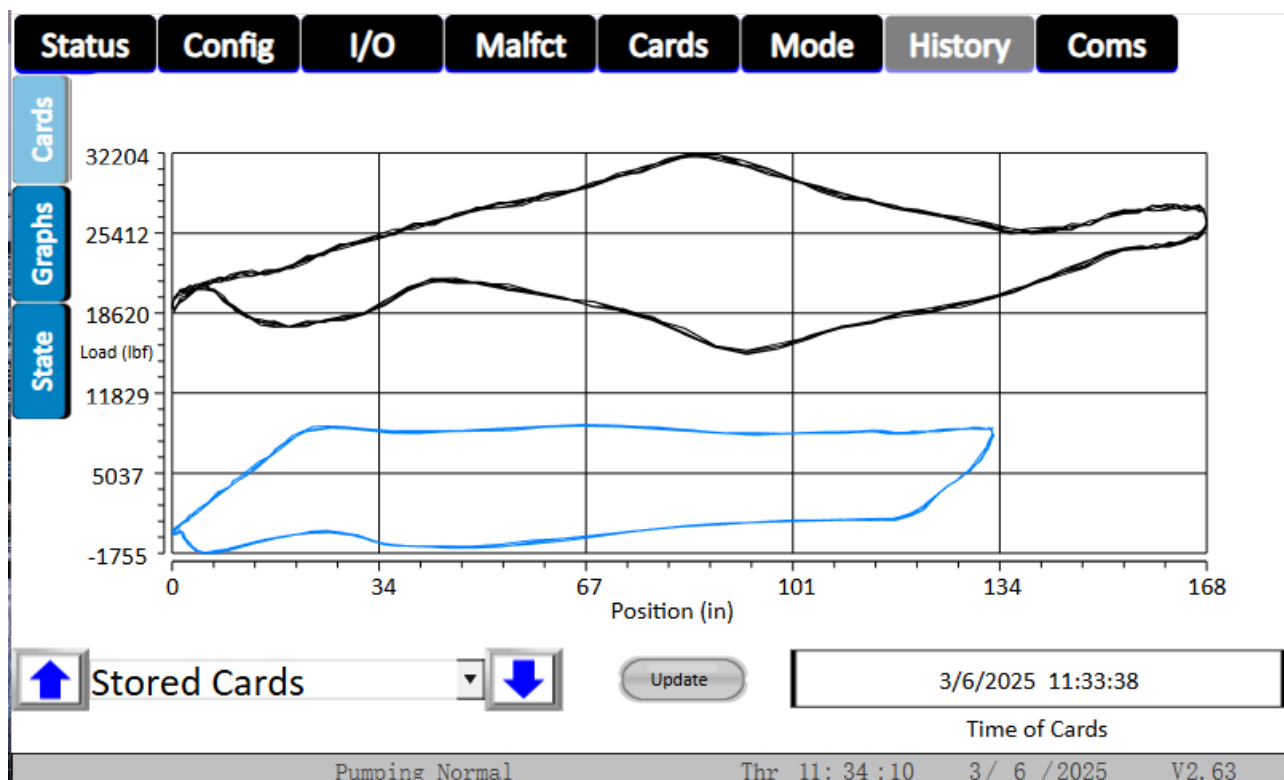


Figure 32 Sample History Cards

History Graph Menu

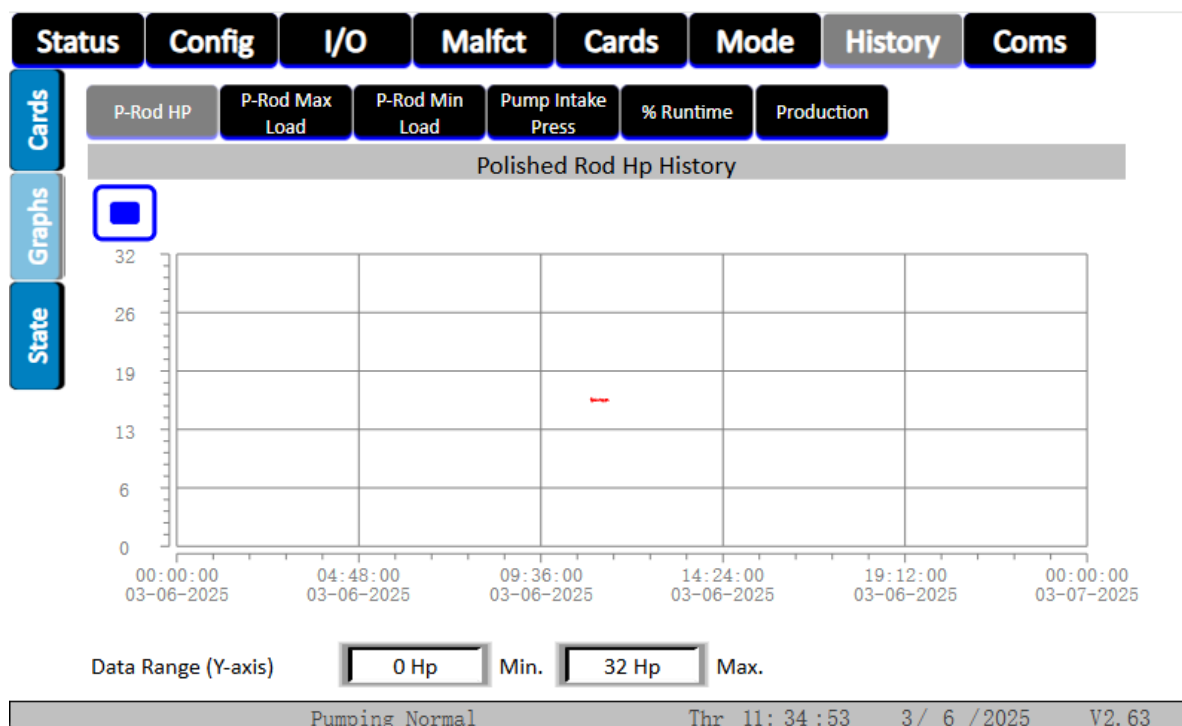


Figure 33 History Graphs Menu

The POC will log the Max. Polish Rod Load, Min. Polish Rod Load, Polish Rod Horsepower, Production, Pump Intake Pressure, and Runtime Percentage once an hour each day and store that data internally. This data can be viewed in one or two ways:

- 1) By visiting one of the history graph pages from the *History Graph Menu* page or
- 2) By viewing the CSV files that can be stored to a USB flash-drive from a PC.

To copy the CSV files from the POC to the flash-drive, insert the flash-drive into the USB port on the front of the cabinet, wait approximately 10 seconds and then press the **Transfer Data Record to USB Device** button on the *History Graph Menu* page. A small, embedded window should appear briefly. Do not touch anything on this window, but instead let it complete its tasks and disappear. Once the window has disappeared, wait another 10 seconds then remove the USB flash drive and insert it into an available USB port on a PC. Once the drive has been recognized, the user can navigate the folder hierarchy and locate the appropriate log files using a file explorer tool, such as Windows Explorer. Open the desired CSV with any spreadsheet program capable of handling CSVs and view the data.

Viewing the data using the history graphs will be covered in the section below titled *Graph Example*, while the *Polish Rod Load vs Time* plot will be discussed in the section with that name following the *Graph Example* section.

Notes:

- 1) The recommended size for a USB flash drive is 4 Gigabytes (GB) or less. The HMI may not correctly write files to the flash drive if the drive has a storage capacity greater than 4 GB.
- 2) When the logger files have been stored on a flash drive, they are stored under the following directory – Flash Drive/DL/1/Year/ (e.g. (G:)/DL/1/2016/ would contain all the log files for the year 2016)

Graph Example

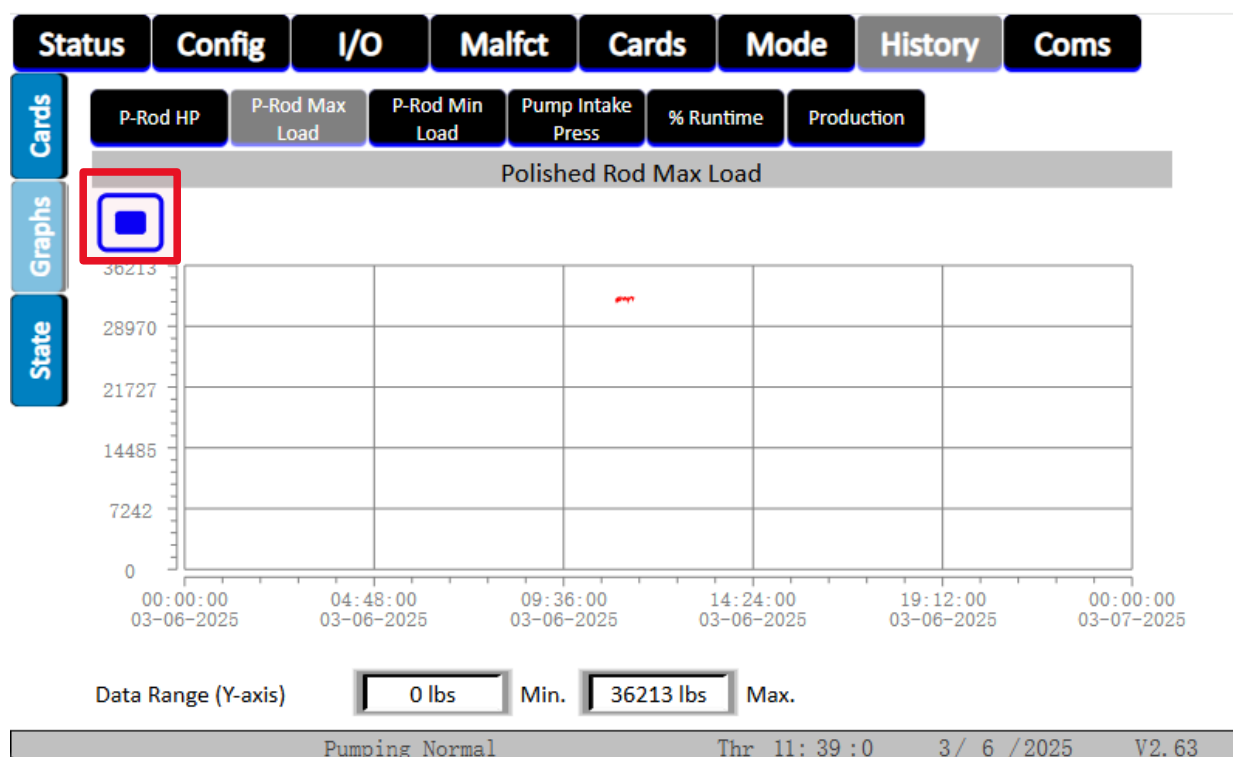


Figure 34 Example Graph

There are six history graphs that are viewable by the user and all of them are set up using the same procedure. With that said, only one of these pages will be covered, but the information presented is applicable to all the graph pages.

To begin, the graphs can be set to their default settings by pressing the **Restore Factory Settings** button on the Pumping Unit Selection page (although pressing this button will also reset several other parameters of the POC and is therefore not recommended when the graphs are the only group that needs to be reset to their defaults).

By pressing the blue square in the top left-hand corner of the graph you can access the control of the data range (Y-Axis) of the graph. Control of the graph is split up into three different pieces – **Data Range**, **Window Time Span** and **Curve Start Time**. The first and easiest piece to understand is the **Data Range** Min./Max. controls. These controls change the range of data that will be displayed along the Y-axis. For example, placing the minimum equal to 100 and the maximum equal to 5000 will cause only values greater than or equal to 100 and less than or equal to 5000 to be displayed in the graph, with all other values being displayed as either 100, if it is below the minimum, or 5000, if it is above the maximum.

The **Window Time Span** is used to set the amount of data that will be visible along the X-axis of the graph, in a unit of time. As an example, setting this span to 99 Days, 0 Hours, 0 Minutes, and 0 Seconds will cause the graph to display 99 days' worth of data, while setting the span to 0 days, 5 hours, 30 minutes, and 10 seconds will cause the graph to zoom in on the data set.

Working hand-in-hand with the Window Time Span is the **Curve Start Time**. The controls within this group set the X-axis offset for the X-axis window (aka the Window Time Span). Adjusting the time within these controls will move the X-axis window forward in time (by setting the time to a value greater than its current time) or backward in time (by setting the time to a value less than the current time). For example, if the user wanted to view 5 hours' worth of data starting at 2:35 p.m. on Oct. 11th, 2016, within the Curve Start Time controls, they would set the Month equal to 10, Day equal to 11, Year equal to 2016, Hour equal to 14 (because this is based on a 24-hour clock), Minute equal to 35, and Second equal to 0 while setting the Window Time Span controls equal to 0 Days, 5 Hours, 0 Minutes, and 0 Seconds.

Notes:

- 1) The **Curve Start Time** is updated automatically once an hour and is set the current time minus the total amount of time set in the **Window Time Span** controls.
- 2) The data displayed on the *Min.* and *Max. Polish Rod Load* pages has been divided by 10 for display purposes (e.g. a value of 3000.0 is a value of 30000). Please keep this in mind when viewing the data and setting the **Data Range** limits on these two pages.

Alarms

The Alarms page contains several elements for keeping track of the state of the pumping unit. The list below provides a description for each of these elements and the figure below displays an example of this page.

- **Well State** – This indicator displays the current state of the POC.
- **Alarm History Display** – This display logs the state of the POC with the time it entered each state on the far-left side of the display, followed by the time it exited each state and the name of each state at the end of each row. NOTE: On the current implementation, the time exiting a state will be approximately the same as the time the state has been entered.
- **RESET** – Pressing this button will clear any active fault on the POC.
- **Time in State** – This indicator shows the amount of time that the POC has been in the current state displayed in the Well State indicator.

Status

Config

I/O

Malfct

Cards

Mode

History

Coms

Cards

Graphs

State

OccurrenceTime	AlarmText
2025-03-06 10:54:27	Pumping Normal

Figure 35 Alarms page



Figure 36 VFD HMI

VFD HMI will display and VFD specific alarms/faults and annunciate all well shutdowns. VFD specific alarms will need to be reset via red push button on the VFD HMI.

Communication Port Settings

The Communications Port Settings page is used to set up the necessary POC parameters to allow for communication between the POC and an external device using the MODBUS Protocol in RTU mode, with the POC acting as the slave and the external device acting as the master. Each column on the page represents settings for a specific communications port, with the name of each port designated at the top of the column. The fields, which are common for each of the ports, are described below with an example shown in figure following the parameter descriptions. **NOTE: If a Wi-Fi module is installed, it uses communications port 0 with the following settings – RTU Address set to 1, Baud Rate set to 115200, Data Bits set to 8, Stop Bits set to 1, and Parity set to None. NOTE: Ensure that the Com Port Received Bytes that you've selected is counting by 8 if it's not in 30 seconds reverse polarity on wiring.**

- **RTU Address** – This is the slave ID number for the POC, which can be set by pressing the field and entering a number between 1 and 247. If an address from 248 to 2295 is entered the system automatically switches from standard Modbus protocol to ELAM (Extended Lufkin Automation Modbus) protocol.
- **Baud Rate (bps)** – The options for this dropdown menu are 300, 1200, 2400, 4800, 9600, 19200, 38400, 57600, and 115200 bits per second.
- **Data Bits** - The options for this dropdown menu are 5, 6, 7, or 8 bits.
- **Stop Bits** - The options for this dropdown menu are 1, 2, or 1.5 bits.
- **Parity** - The options for this dropdown menu are None, Even, or Odd.
- **Received Bytes** – This is the number of received bytes since last reset or clear.
- **Com 3** – is dedicated for HMI communication from the POC board.

Status					Config					I/O					Malft					Cards					Mode					History					Coms																								
Serial					Com Port 0 Settings					Com Port 1 Settings					Com Port 2 Settings																																												
Port Enable					ENABLED					ENABLED					ENABLED																																												
RTU Address					1					1					1																																												
Baud Rate (bps)					115200					115200					115200																																												
Data Bits					8					8					8																																												
Stop Bits					1					1					1																																												
Parity					None					None					None																																												
Received Bytes					0					0					174815																																												
					Com 3 Received Bytes					137126438																																																	
Pumping Normal																														Thr 11: 44 :33										3 / 6 /2025										V2.63									

Figure 37 Communication Port Settings

Communication HMI Ethernet

The Coms HMI Ethernet page allows the user to configure the HMI's ethernet IP settings to their desired address. To edit the local IP settings press and hold the SYS button in the top right-hand corner until the screen changes. After the system screen is displayed select network, then select ethernet and change settings as needed. When done press the restart button. **It's important that the unit be stopped and in the Off position when doing this to avoid any system malfunctions.** After making the necessary IP changes the user must confirm the IP change and reboot the HMI via the "REBOOT" button in orange.

StatusConfigI/OMalfctCardsModeHistoryComsSYS

SerialHMI EthPOC Eth

To Change Local IP settings press and hold SYS button above until screen change

After system screen is displayed select Network
Select Ethernet and change settings as needed.
When Done Press the Restart Button

IP Mode
Static▲

IP	192	168	1	91
Netmask	255	255	255	0
Gateway	192	168	1	1
DNS1	192	168	1	1
DNS2	0	0	0	0

Confirm IP ChangeREBOOT

Pumping NormalThr 11:45:223 / 6 / 2025V2.63

Figure 38 Communication HMI Ethernet

Communication POC Ethernet

The Coms POC Ethernet page allows the user to configure the HMI's ethernet IP settings to their desired address. **It's important that the unit be stopped and in the Off position when doing this to avoid any system malfunctions.** After making the necessary IP changes the user must confirm the IP change and reboot the HMI via the "REBOOT" button in orange.

IP	192	168	1	29
Netmask	255	255	255	0
Gateway	192	168	1	1
DNS1	0	0	0	0
DNS2	0	0	0	0

REBOOT POC

Pumping Normal Thr 11:46:14 3 / 6 / 2025 V2.63

Figure 39 Communication HMI Ethernet

Wireless Load Cell Configuration

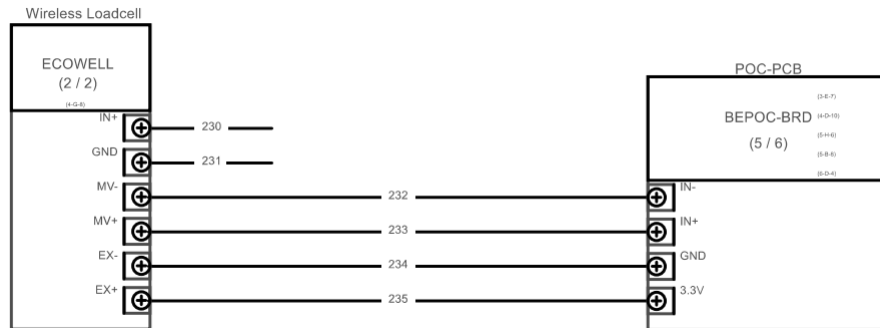


Figure 40 Wireless Loadcell Connection

Place the spare battery into the WS unit, then mount the WS unit to polish rod. Unscrew the side door nut, plug in the spare battery connector, the red wire is on the top and the black wire is on the bottom, put the battery in, tighten the nut.

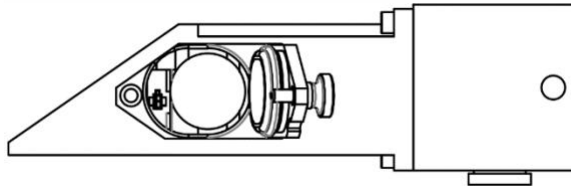


Figure 41 Wireless Battery Hatch

Install the arc-shaped block and then insert the screw to avoid wear between the polished rod and the screw.

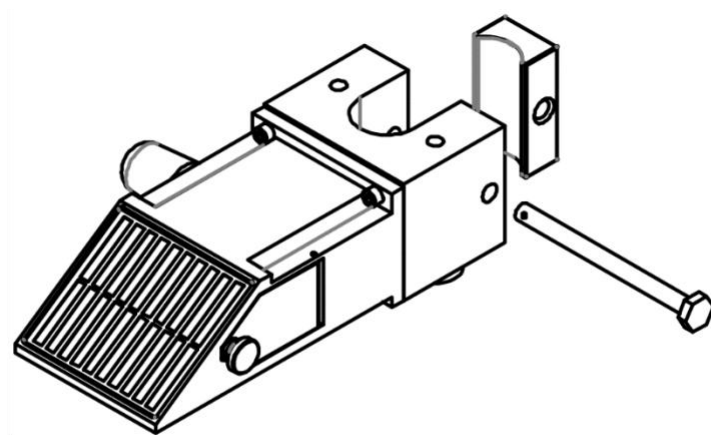
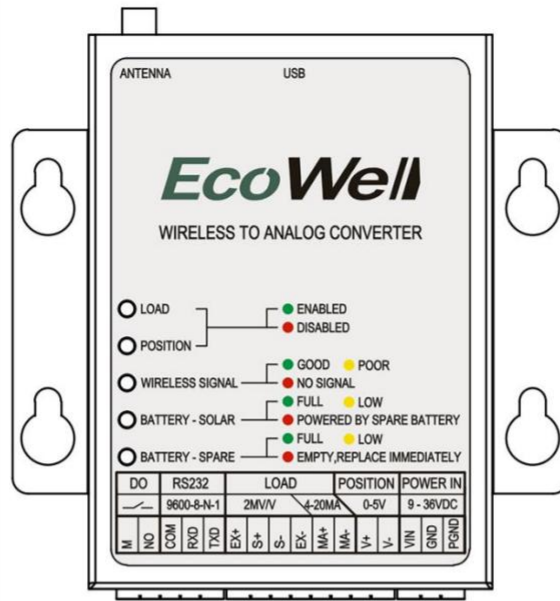


Figure 41a Example Load Cell Install

Figure 42 Ecowell Diagram



Terminal	Specification	Signal	Description	Function
POWER IN	9-36VDC	VIN	Power input +	Power input for WAC unit
		GND	Power input -	
		PGND	Connect to Earth Ground	
POSITION	0-5V	V+	Position voltage signal +	True Position signal output (Not Acceleration signal)
		V-	Position voltage signal -	
LOAD	4-20MA	MA+	Load mA signal +	Load mA signal output
		MA-	Load mA signal -	
	2MV/V	EX+	Excitation Voltage +	Load mV signal output EX=5V/8V/10V S=EX*2MV/V
		S+	Load mV signal +	
		S-	Load mV signal -	
		EX-	Excitation Voltage -	
RS232	9600-8-N-1	COM	Common terminal	RS232 communicate interface
		RXD	Receive Data	
		TXD	Transmit Data	
DO	RELAY OUTPUT	M	Common terminal	Spare Battery status indicate Open: Battery FULL or LOW Close: Battery EMPTY
		NO	Normal Open	

Wired Load Cell Configuration

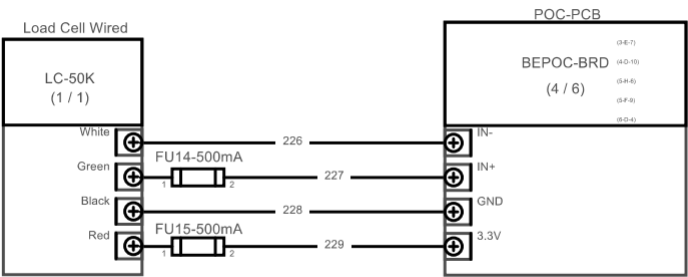


Figure 43 Wired Loadcell Connection

Hall Effects Kit Installation

Installing the Polished Rod Load Cell

The polished rod load cell (PRLC) is installed between the polished rod clamp and the carrier bar as shown in Figure A. A spherical washer set must be installed between the carrier bar and the PRLC to ensure an accurate load signal. When the load cell is properly installed, the load cell supports the full load of the rod string. The PRLC provides a quantitative signal that is directly proportional to the load changes occurring at the polished rod.

For all wired load cell installations, a spherical washer set must be placed between the top of the carrier bar and the bottom flange of the load cell.

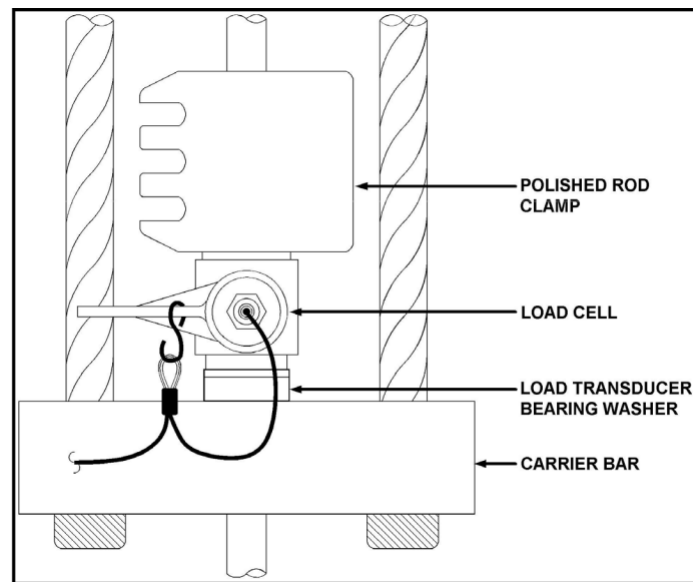


Figure 44 Polish Rod Load Cell Installed with spherical washer set

Note: If the rod rotator is installed, ensure the load cell is installed below the rod rotator, or severe damage can occur to the load cell and/or cable.

Installation Procedure for Reference Only

Follow the steps below to properly install the load cell.

1. Stop the pumping unit on the downstroke.

2. Create a space between the carrier bar and the polished rod clamp for placing the load cell by stacking out the rod string on the wellhead (i.e., temporarily set the rod string on the wellhead). This can be done using a winch truck or by using the pumping unit prime mover. The following steps are suggested:

a. Make sure the unit brake is holding securely.

b. Install a temporary polished rod clamp on the polished rod below the carrier bar and above the polished rod liner using the following precautions:

- Use the correct clamp size.
- Tighten the clamp securely and use a handle extension as necessary. Use double-bolt clamps in deep wells with heavy rod strings.
- On hard-faced polished rods, attempt to install the clamp on a portion of the polished rod that is not hard-faced.

c. Protect the stuffing box using the following precautions:

- Do not set down the weight of the rod string on the stuffing box packing. Use the proper stuffing box protectors.
- Select the proper standoff length to avoid throwing too much slack in the bridge.
- Throw slack at the carrier bar as gently as possible. Avoid “running” into the standoff.

Note: Installing the PRLC often requires climbing on the wellhead or using a ladder. Use extreme caution when climbing using a ladder. Set it on level ground. Some companies prohibit climbing on the wellhead.

DANGER: Keep fingers and hands out of harm’s way in case the polished rod clamp or unit brake slips.

3. After the rod string is independently supported, mark the position of the original rod clamp on the polished rod with a grease pencil or soapstone.

4. Turn off all power at the main supply. Use a voltmeter to ensure that no power is coming to the motor starter panel.

5. Turn off the RPC. If it has an optional battery, it may have the RPC powered up even though the AC is no longer on.

6. Remove the original polished rod clamp from above the carrier bar.

7. Remove the pony rod or collar from the top of the polished rod. 8. Slide the spherical washer set down over the polished rod and position it on top of the carrier bar.

9. Place the load cell around the polished rod on top of the spherical washer set. Observe the Up arrow on the side of the PRLC and be sure that it points up.

10. Check that the load-bearing surface of the PRLC rests on the surface of the spherical washer set.

11. Position the PRLC so that the signal wire socket faces the pumping unit.

Note: The PRLC must be installed below the rod rotator if one is present, or severe damage can occur to the load cell and/or cable.

12. Replace the pony rod or collar with the polished rod.

13. Replace the original rod clamp above the PRLC. Remember to move the clamp up from the original marked position by the height of the load cell and spherical washer set so that the same downhole pump spacing is maintained.

14. GENTLY release the unit brake to transfer the weight of the rod string back to the carrier bar and bridle. DO NOT drop the rod string abruptly onto the load cell. A shock loading can introduce a zero offset in the transducer, making the load cell calibration ineffective. Reset the brake after obtaining a small amount of standoff clearance. Remove the standoff and the temporary polished rod clamp installed below the carrier bar.

15. Operate the unit slowly during the first pump cycle to check for possible problems, such as

- Bridle slipping off the horsehead due to poor horsehead alignment
- Polished rod liner (if installed) being pulled out of the stuffing box packing, which will create a hazardous leak
- Inadequate clearance between the horsehead and polished rod clamp when the unit is at the top of stroke.

The PRLC is now installed.

Installing the Motor RPM Hall-Effect Transducer For Reference Only

The magnet assembly must be mounted on the motor shaft and the Hall-Effect transducer probe must be mounted so that the magnet passes within about 1/8 inch of the probe tip when it passes the probe. The installation procedure provided below assumes that the belt sheave is far enough out on the motor shaft to mount the magnet between the motor and the sheave. This will work for most installations. The installation technician may need to be innovative in some locations.

Tools Required

- Socket set or wrench set
- Screwdriver

Parts Required

- Transducer assembly with motor bracket and T-bracket
- Magnet assembly
- Hose clamp

Installation Procedure

Follow the steps below to properly install the motor RPM Hall-Effect transducer.

- 1 Turn off the main power switch to the motor.
- 2 Set the brake on the pumping unit and chain off the unit sheave.
- 3 Attach the magnet (mounted in holder) to the motor shaft with a stainless-steel hose clamp. Cut off excess hose clamp after it is tightened. See Figure 45.

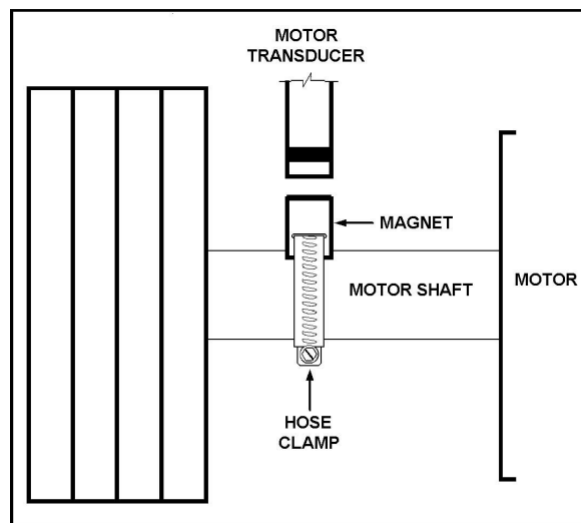


Figure 45 Motor Transducer assembly showing T-Bracket Linkage

- 4 Install the motor transducer bracket on a motor housing bolt. Select a housing bolt that will easily allow the transducer to align with the magnet.

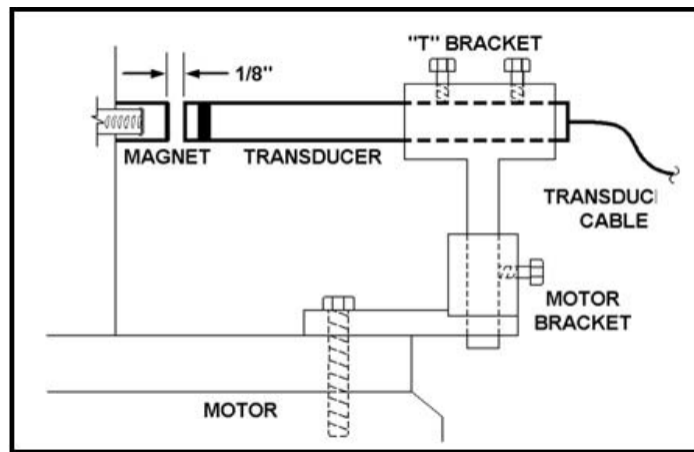


Figure 46 Motor Transducer Assembly

- 5 Assemble and adjust the T-bracket linkage so that the magnet passes approximately 1/8 inch from the transducer. See Figure 46.
- 6 Tighten all bracket connections.

The RPM Hall-Effect transducer is now installed.

Installing the Crank Arm Hall-Effect Transducer For Reference Only

The crank transducer must be precisely installed to sense the leading edge of the magnet at the bottom dead-center of the stroke. An error in this measurement will result in a shift in polished rod positions, which will cause distortions in the surface and downhole pump cards and errors in the diagnostic calculations.

Tools Required

- Socket set or wrench set
- Screwdriver
- Silicon sealant/adhesive

Parts Required

- Transducer assembly and brackets
- Magnet

Installation Procedure

Follow the steps below to properly install the crank arm transducer.

1. Position the unit cranks with the polished rod at bottom-dead-center.
 - a. Align the crank and pitman. The centerlines of the pitman and the crank should be in perfect alignment. See Figure 47.
 - b. Stop the unit near the bottom dead-center.
 - c. With the motor turned off, release the brake slowly and observe the direction that the cranks rotate.
 - d. With this in mind, start the unit and then stop the unit with the polished rod near the bottom. When the brake is released gently, the cranks will rotate slowly in the direction that will align the centerlines of the crank and pitman.

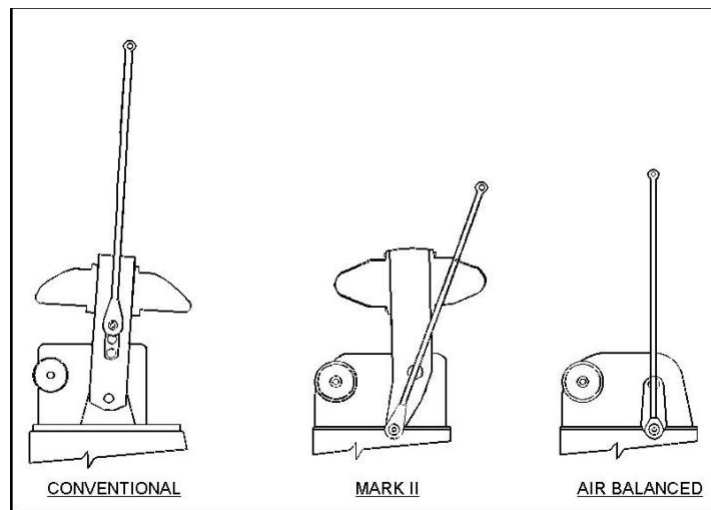


Figure 47 Crank/Pitman Alignment at Bottom Dead-Centre

If one person is involved in the installation, this procedure can be done in steps by allowing the cranks to rotate a small amount, then set the brake and observe the alignment. Continue this method until the centerlines are in perfect alignment. If this method is not successful (unit behavior differs from well to well), consider stopping the unit slightly before bottom-dead-center, then partially set the brake and “bump” the unit by quickly turning the motor on and off. After each “bump,” observe the crank and pitman alignment until you attain perfect alignment.

- 1 Turn off the main power switch to the motor.
 - 2 Set the brake on the pumping unit and chain off the unit sheave.
 - 3 Select a bolt on the gearbox slow speed shaft bearing cap that will allow the easiest and best alignment of the transducer and the magnet. Remove the bolt and mount the bracket of the Hall-Effect transducer. Reinstall and tighten the bolt to secure the bracket. See Figures 48 and 49.
 - 4 Use either method described below to mount a magnet.
- **Bar magnet** (see Figure E) Mount a bar magnet on the crank and carefully inspect the backside of the crank arm to ensure that the transducer will not be hit by a portion of the crank that extends out further than the surface of the magnet. After making sure that the magnet facing the transducer has the correct polarity, glue the magnet to the crank with silicone.

Note: Incorrect polarity will result in no crank signal. Correct polarity can be determined by holding the bar magnet near a round magnet mounted in a holder used for measuring motor revolutions and speed. The side of the bar magnet attracted to the round magnet is the side glued to the crank.

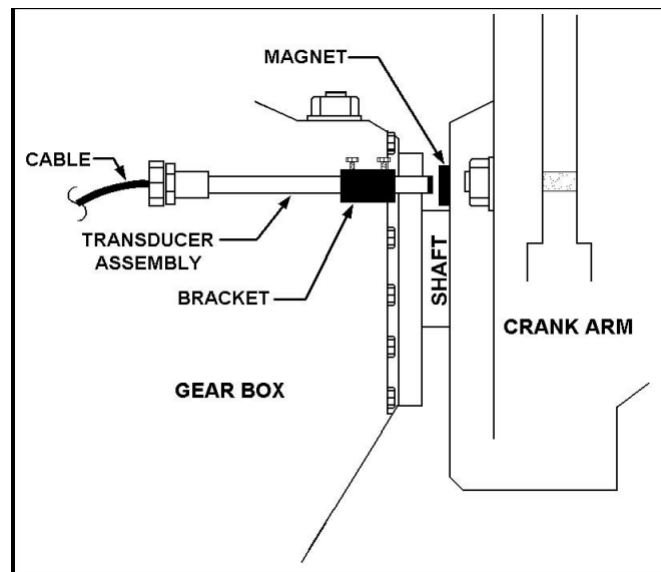


Figure 48 Bar Magnet Mounted on Back of Crank Arm

Round magnet (see Figure 49) — Attach a round magnet (mounted in holder) to the slow speed shaft of the gearbox with a stainless-steel hose clamp, like the motor transducer assembly. If insufficient space is available to mount the magnet between the crank and the gearbox bearing cap, mount a bar magnet on the crank as described above.

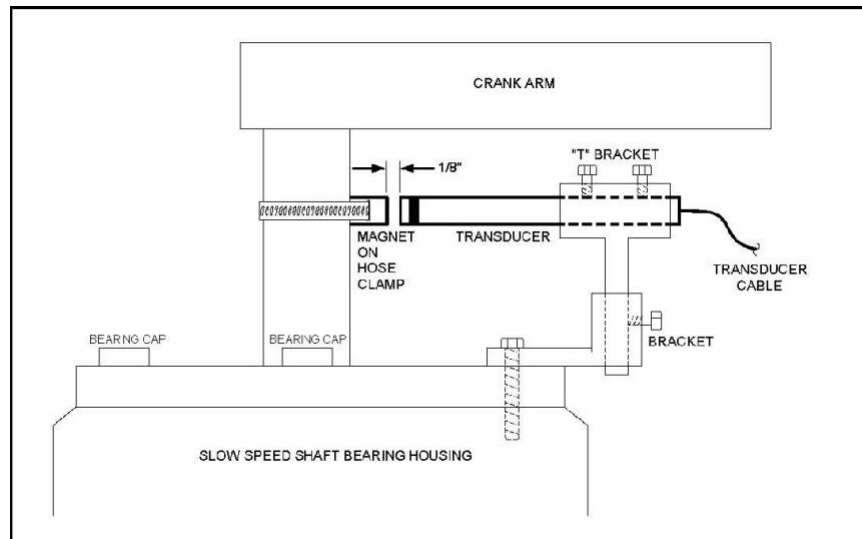


Figure 49 Bar Magnet Mounted on Back of Crank Arm

5 The Hall-Effect transducer senses the magnet when the leading edge of the magnet is approximately at the center of the transducer probe. And with the unit at bottom dead-center, position the transducer or the magnet so that the center of the transducer is at the leading edge of the magnet and approximately 1/8 inch from the magnet (See Figure G for proper magnet alignment.). Tighten all connections securely.

The crank arm Hall-Effect transducer is now installed.

TESTING USE ONLY: Testing of load cell, hall effects kit & position sensors

Testing of load cell & cables before leaving:

- connect sensor cable for position
- test the cable using magnet, ensure CSW light illuminates
- label the cable "position" and leave the cable inside of the VFD package
- connect sensor cable for speed pickup on motor
- test the cable using magnet, ensure SPD light illuminates
- label the cable "motor" and leave the cable inside of the VFD package
- Ensure voltage leaving board on the E+ and E- terminals is 5VDC

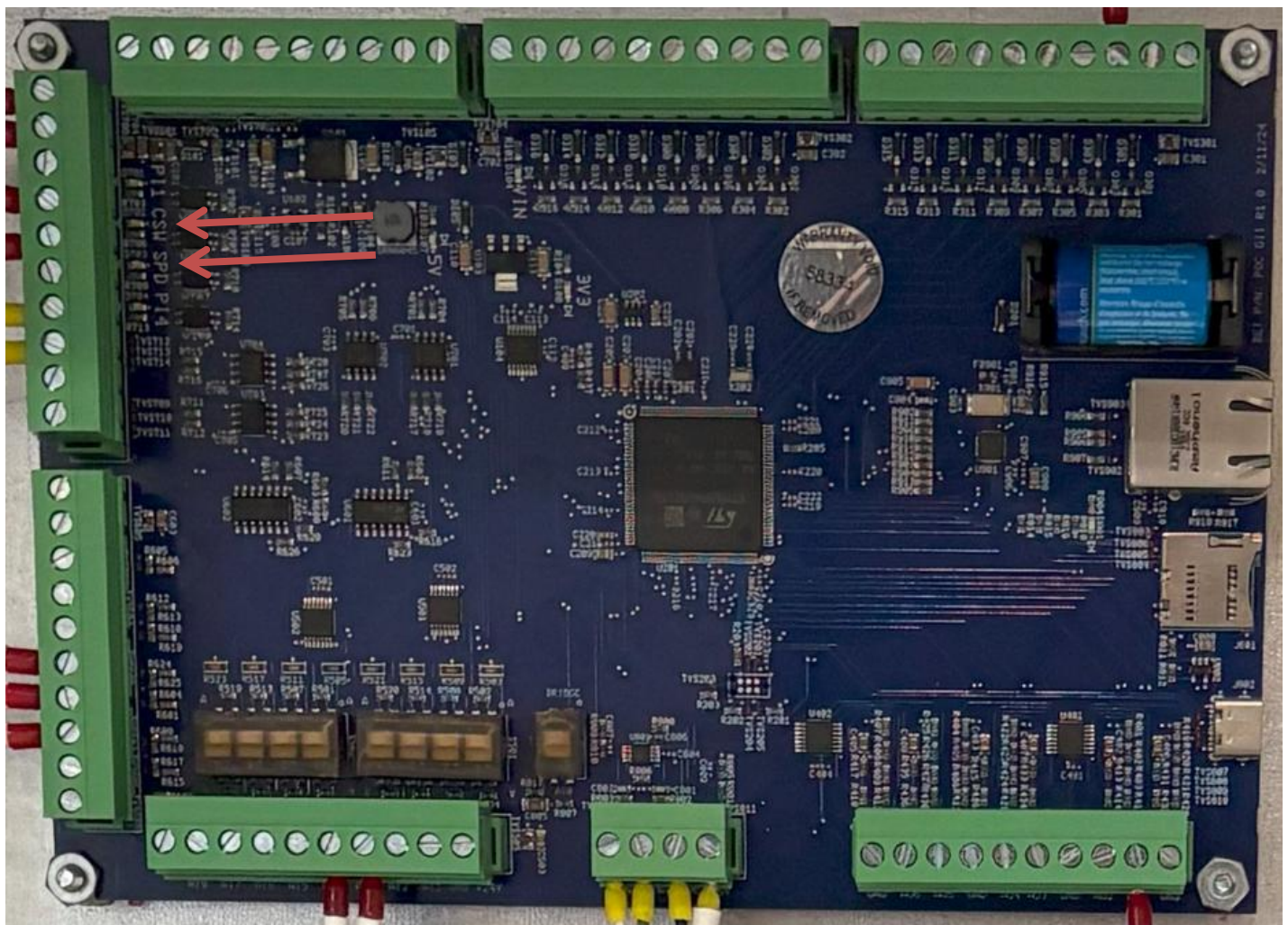


Figure 50 PCB Board

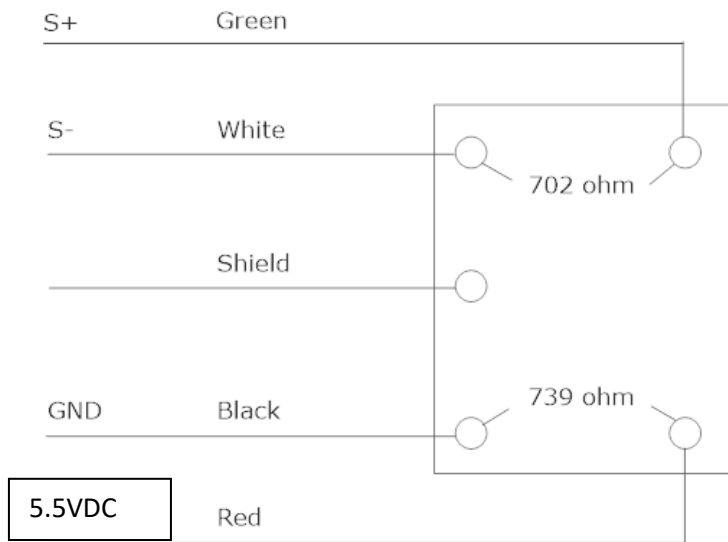


Figure 51 Ohmic Values for wired loadcell

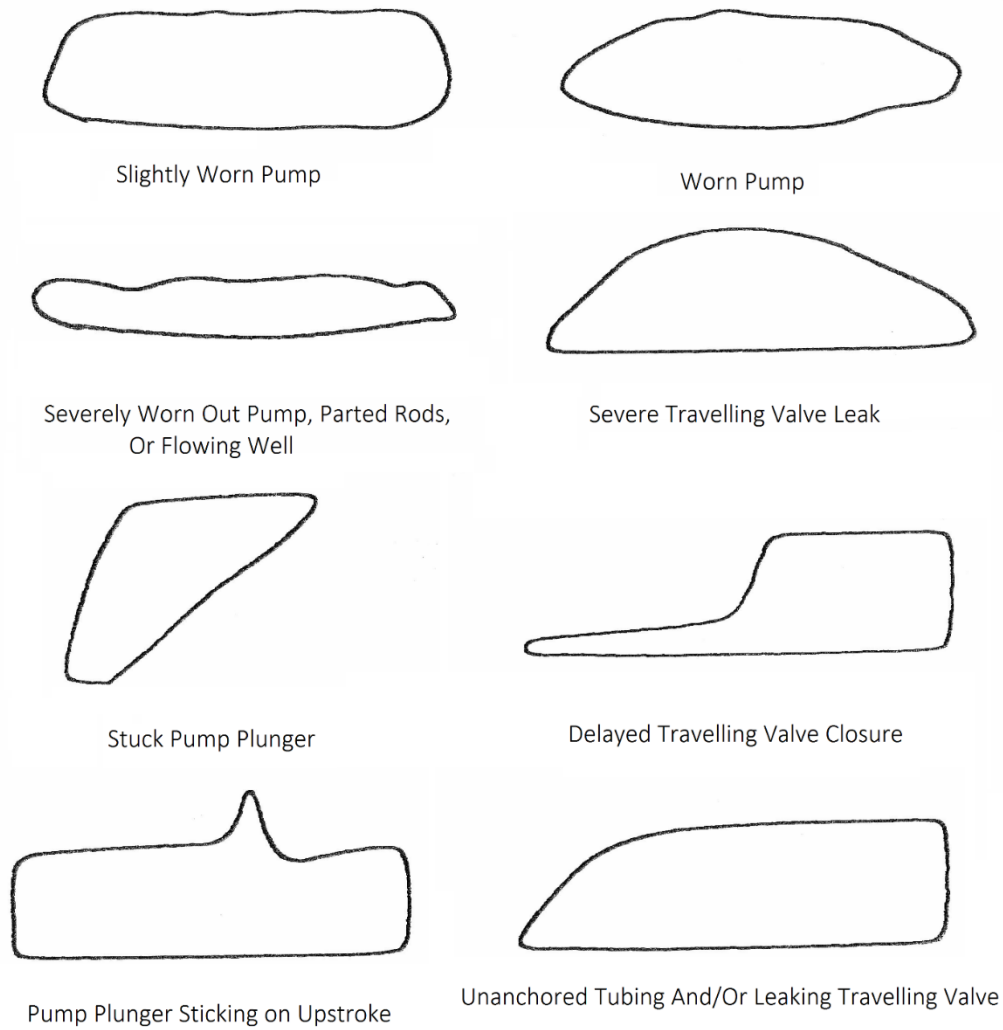
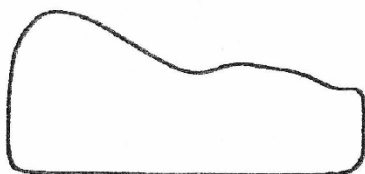
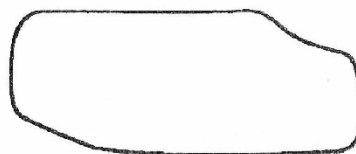


Figure 52 Examples of Pump Cards



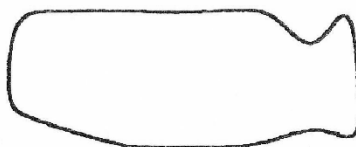
Full Pump - High Fluid Acceleration
(Shallow Well, High Fluid Production)



Full Pump - Shallow Friction.
Tight Stuffing Box, Parafin, etc.



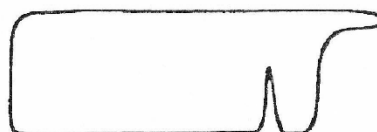
Slight Fluid Pound - Fiberglass Rods



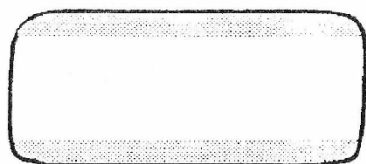
Full Pump - Excessive Stuffing Box Friction



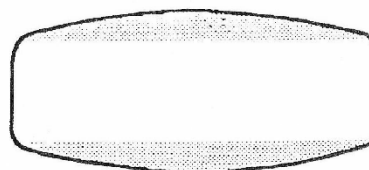
Full Pump - Fiberglass Rods



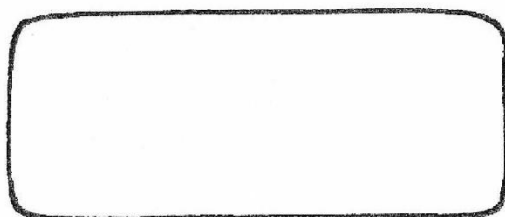
Fluid Pound - Wave Reflection From Sinker Bars



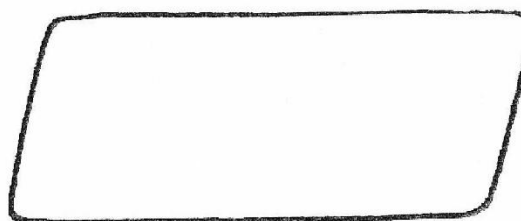
Full Pump - High Rod - On Tubing Friction



Full Pump - High Fluid Friction

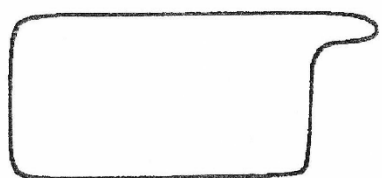


Full Pump - Anchored Tubing



Full Pump - Unanchored Tubing

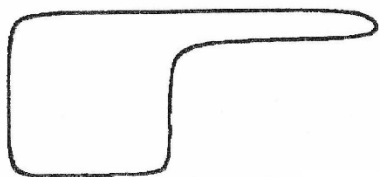
Figure 52-A Examples of Pump Cards



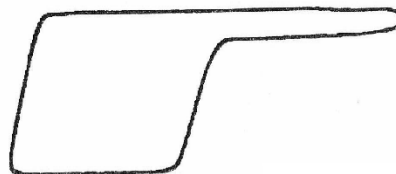
Slight Fluid Pound - Anchored Tubing



Slight Fluid Pound - Unanchored Tubing



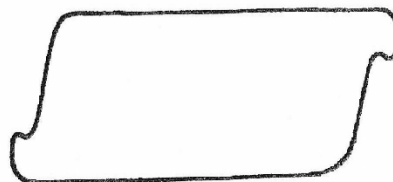
Severe Fluid Pound - Anchored Tubing



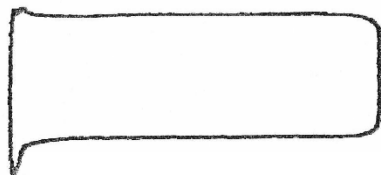
Severe Fluid Pound - Unanchored Tubing



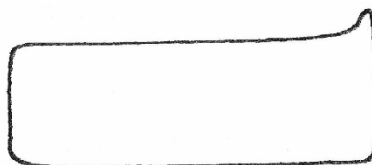
Completely Pumped Off - Anchored Tubing



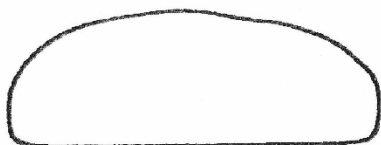
Malfunctioning Tubing Anchor



Full Pump - Pump Tapping Bottom



Full Pump - Pump Tapping Top



Leaking Travelling Valve



Leaking Standing Valve

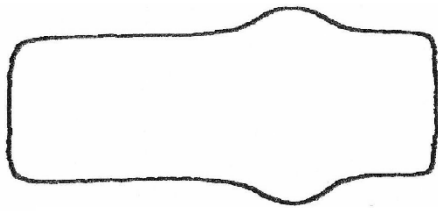


Gas Locked Pump

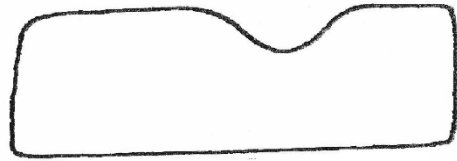


Gas Interference

Figure 52-B Examples of Pump Cards



Bent or Sticking Pump Barrel



Worn Or Split Pump Barrel

Figure 52-B Examples of Pump Cards

Notes:

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